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Three-dimensional data encoding method, three-dimensional data decoding method, three-dimensional data encoding device, and three-dimensional data decoding device

Abstract

A three-dimensional data encoding method includes: quantizing geometry information of each of three-dimensional points, using a first quantization parameter; quantizing a first luminance using a second quantization parameter and quantizing a first chrominance using a third quantization parameter, the first luminance and the first chrominance indicating a first color among attribute information of each of the three dimensional points; and generating a bitstream including the geometry information quantized, the first luminance quantized, the first chrominance quantized, the first quantization parameter, the second quantization parameter, and a first difference between the second quantization parameter and the third quantization parameter.

Inventors: Han; Chung Dean (Singapore, SG), Lasang; Pongsak (Singapore, SG), Wang; Chi (Singapore, SG), Iguchi; Noritaka (Osaka, JP), Sugio; Toshiyasu (Osaka, JP)

Applicant: Panasonic Intellectual Property Corporation of America (Torrance, CA)

Family ID: 1000008781262

Assignee: PANASONIC INTELLECTUAL PROPERTY CORPORATION OF AMERICA (Torrance, CA)

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Primary Examiner: Hasan; Mainul

Attorney, Agent or Firm: Wenderoth, Lind & Ponack, L.L.P.

Background/Summary

BACKGROUND

1. Technical Field

(1) The present disclosure relates to a three-dimensional data encoding method, a three-dimensional data decoding method, a three-dimensional data encoding device, and a three-dimensional data decoding device.

2. Description of the Related Art

(2) Devices or services utilizing three-dimensional data are expected to find their widespread use in a wide range of fields, such as computer vision that enables autonomous operations of cars or robots, map information, monitoring, infrastructure inspection, and video distribution. Three-dimensional data is obtained through various means including a distance sensor such as a rangefinder, as well as a stereo camera and a combination of a plurality of monocular cameras.

(3) Methods of representing three-dimensional data include a method known as a point cloud scheme that represents the shape of a three-dimensional structure by a point cloud in a three-dimensional space. In the point cloud scheme, the positions and colors of a point cloud are stored. While point cloud is expected to be a mainstream method of representing three-dimensional data, a massive amount of data of a point cloud necessitates compression of the amount of three-dimensional data by encoding for accumulation and transmission, as in the case of a two-dimensional moving picture (examples include Moving Picture Experts Group-4 Advanced Video Coding (MPEG-4 AVC) and High Efficiency Video Coding (HEVC) standardized by MPEG).

(4) Meanwhile, point cloud compression is partially supported by, for example, an open-source library (Point Cloud Library) for point cloud-related processing.

(5) Furthermore, a technique for searching for and displaying a facility located in the surroundings of the vehicle by using three-dimensional map data is known (for example, see International Publication WO 2014/020663).

SUMMARY

(6) There has been a demand for improving coding efficiency in encoding and decoding three-dimensional data.

(7) The present disclosure provides a three-dimensional data encoding method, a three-dimensional data decoding method, a three-dimensional data encoding device, or a three-dimensional data decoding device that is capable of improving coding efficiency.

(8) A three-dimensional data encoding method according to an aspect of the present disclosure includes: quantizing geometry information of each of three-dimensional points, using a first quantization parameter; quantizing a first luminance using a second quantization parameter and quantizing a first chrominance using a third quantization parameter, the first luminance and the first chrominance indicating a first color among attribute information of each of the three dimensional points; and generating a bitstream including the geometry information quantized, the first luminance quantized, the first chrominance quantized, the first quantization parameter, the second quantization parameter, and a first difference between the second quantization parameter and the third quantization parameter.

(9) A three-dimensional data decoding method according to an aspect of the present disclosure includes: obtaining quantized geometry information, quantized first luminance, quantized first chrominance, a first quantization parameter, a second quantization parameter, and a first difference between the second quantization parameter and a third quantization parameter, by obtaining a bitstream; calculating geometry information of three-dimensional points by inverse-quantizing the quantized geometry information using the first quantization information; calculating a first luminance, out of the first luminance and a first chrominance which indicate a first color of the three-dimensional points, by inverse-quantizing the quantized first luminance using the second quantization parameter; and calculating the first chrominance by inverse-quantizing the quantized first chrominance using the third quantization parameter obtained from the second quantization parameter and the first difference.

(10) The present disclosure can provide a three-dimensional data encoding method, a three-dimensional data decoding method, a three-dimensional data encoding device, or a three-dimensional data decoding device that is capable of improving coding efficiency.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) These and other objects, advantages and features of the disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

- (2) FIG. 1 is a diagram illustrating a configuration of a three-dimensional data encoding and decoding system according to Embodiment 1;
- (3) FIG. 2 is a diagram illustrating a structure example of point cloud data according to Embodiment 1;
- (4) FIG. 3 is a diagram illustrating a structure example of a data file indicating the point cloud data according to Embodiment 1;
- (5) FIG. 4 is a diagram illustrating types of the point cloud data according to Embodiment 1;
- (6) FIG. 5 is a diagram illustrating a structure of a first encoder according to Embodiment 1;
- (7) FIG. 6 is a block diagram illustrating the first encoder according to Embodiment 1;
- (8) FIG. 7 is a diagram illustrating a structure of a first decoder according to Embodiment 1;
- (9) FIG. 8 is a block diagram illustrating the first decoder according to Embodiment 1;
- (10) FIG. 9 is a diagram illustrating a structure of a second encoder according to Embodiment 1;
- (11) FIG. 10 is a block diagram illustrating the second encoder according to Embodiment 1;
- (12) FIG. 11 is a diagram illustrating a structure of a second decoder according to Embodiment 1;
- (13) FIG. 12 is a block diagram illustrating the second decoder according to Embodiment 1;
- (14) FIG. 13 is a diagram illustrating a protocol stack related to PCC encoded data according to Embodiment 1;
- (15) FIG. 14 is a block diagram of an encoder according to Embodiment 1;
- (16) FIG. 15 is a block diagram of a decoder according to Embodiment 1;
- (17) FIG. 16 is a flowchart of encoding processing according to Embodiment 1;
- (18) FIG. 17 is a flowchart of decoding processing according to Embodiment 1;
- (19) FIG. 18 is a diagram illustrating a basic structure of ISOBMFF according to Embodiment 2;
- (20) FIG. 19 is a diagram illustrating a protocol stack according to Embodiment 2;
- (21) FIG. 20 is a diagram illustrating an example where a NAL unit is stored in a file for codec 1 according to Embodiment 2;
- (22) FIG. 21 is a diagram illustrating an example where a NAL unit is stored in a file for codec 2 according to Embodiment 2;
- (23) FIG. 22 is a diagram illustrating a structure of a first multiplexer according to Embodiment 2;
- (24) FIG. 23 is a diagram illustrating a structure of a first demultiplexer according to Embodiment 2;
- (25) FIG. 24 is a diagram illustrating a structure of a second multiplexer according to Embodiment 2;
- (26) FIG. 25 is a diagram illustrating a structure of a second demultiplexer according to Embodiment 2;
- (27) FIG. 26 is a flowchart of processing performed by the first multiplexer according to Embodiment 2;
- (28) FIG. 27 is a flowchart of processing performed by the second multiplexer according to Embodiment 2;
- (29) FIG. 28 is a flowchart of processing performed by the first demultiplexer and the first decoder according to Embodiment 2;
- (30) FIG. 29 is a flowchart of processing performed by the second demultiplexer and the second decoder according to Embodiment 2;
- (31) FIG. 30 is a diagram illustrating structures of an encoder and a third multiplexer according to Embodiment 3;
- (32) FIG. 31 is a diagram illustrating structures of a third demultiplexer and a decoder according to Embodiment 3;
- (33) FIG. 32 is a flowchart of processing performed by the third multiplexer according to Embodiment 3;
- (34) FIG. 33 is a flowchart of processing performed by the third demultiplexer and the decoder according to Embodiment 3;

- (35) FIG. 34 is a flowchart of processing performed by a three-dimensional data storage device according to Embodiment 3;
- (36) FIG. 35 is a flowchart of processing performed by a three-dimensional data acquisition device according to Embodiment 3;
- (37) FIG. 36 is a diagram illustrating structures of an encoder and a multiplexer according to Embodiment 4;
- (38) FIG. 37 is a diagram illustrating a structure example of encoded data according to Embodiment 4;
- (39) FIG. 38 is a diagram illustrating a structure example of encoded data and a NAL unit according to Embodiment 4;
- (40) FIG. 39 is a diagram illustrating a semantics example of pcc_nal_unit_type according to Embodiment 4;
- (41) FIG. 40 is a diagram illustrating an example of a transmitting order of NAL units according to Embodiment 4;
- (42) FIG. 41 is a flowchart of processing performed by a three-dimensional data encoding device according to Embodiment 4;
- (43) FIG. 42 is a flowchart of processing performed by a three-dimensional data decoding device according to Embodiment 4;
- (44) FIG. 43 is a flowchart of multiplexing processing according to Embodiment 4;
- (45) FIG. 44 is a flowchart of demultiplexing processing according to Embodiment 4;
- (46) FIG. 45 is a flowchart of processing performed by a three-dimensional data encoding device according to Embodiment 4;
- (47) FIG. 46 is a flowchart of processing performed by a three-dimensional data decoding device according to Embodiment 4;
- (48) FIG. 47 is a block diagram illustrating a divider according to Embodiment 5.
- (49) FIG. 48 is a diagram illustrating an example of slice division and tile division according to Embodiment 5.
- (50) FIG. 49 is a diagram illustrating an example of a slice division pattern and a tile division pattern according to Embodiment 5.
- (51) FIG. 50 is a flowchart of a process of re-initializing a CABAC encoding/decoding engine in response to a CABAC initialization flag in encoding or decoding according to Embodiment 6.
- (52) FIG. 51 is a block diagram illustrating a configuration of first encoder included in a three-dimensional data encoding device according to Embodiment 6.
- (53) FIG. 52 is a block diagram illustrating a configuration of a divider according to Embodiment 6.
- (54) FIG. 53 is a block diagram illustrating a configuration of a geometry information encoder and an attribute information encoder according to Embodiment 6.
- (55) FIG. 54 is a block diagram illustrating a configuration of a first decoder according to Embodiment 6.
- (56) FIG. 55 is a block diagram illustrating a configuration of a geometry information decoder and an attribute information decoder according to Embodiment 6.
- (57) FIG. 56 is a flowchart illustrating an example of a process associated with the initialization of CABAC in the encoding of geometry information or the encoding of attribute information according to Embodiment 6.
- (58) FIG. 57 is a diagram illustrating an example of timings of CABAC initialization for point cloud data in the form of a bitstream according to Embodiment 6.
- (59) FIG. 58 is a diagram illustrating a configuration of encoded data and a method of storing the encoded data into a NAL unit according to Embodiment 6.
- (60) FIG. 59 is a flowchart illustrating an example of a process associated with the initialization of CABAC in the decoding of geometry information or the decoding of attribute information

according to Embodiment 6.

(61) FIG. **60** is a flowchart of a process of encoding point cloud data according to Embodiment 6.

(62) FIG. **61** is a flowchart illustrating an example of a process of updating additional information according to Embodiment 6.

(63) FIG. **62** is a flowchart illustrating an example of a process of initializing CABAC according to Embodiment 6.

(64) FIG. **63** is a flowchart illustrating a process of decoding point cloud data according to Embodiment 6.

(65) FIG. **64** is a flowchart illustrating an example of a process of initializing a CABAC decoder according to Embodiment 6.

(66) FIG. **65** is a diagram illustrating an example of tiles and slices according to Embodiment 6.

(67) FIG. **66** is a flowchart illustrating an example of a method of determining whether to initialize CABAC and determining a context initial value according to Embodiment 6.

(68) FIG. **67** is a diagram illustrating an example of a case where a map, which is a top view of point cloud data obtained by LiDAR, is divided into tiles according to Embodiment 6.

(69) FIG. **68** is a flowchart illustrating another example of the method of determining whether to initialize CABAC and determining a context initial value according to Embodiment 6.

(70) FIG. **69** is a diagram for describing a process performed by a quantizer and an inverse quantizer according to Embodiment 7.

(71) FIG. **70** is a diagram for describing a default value and a quantization delta of a quantization value according to Embodiment 7.

(72) FIG. **71** is a block diagram illustrating a configuration of a first encoder included in a three-dimensional data encoding device according to Embodiment 7.

(73) FIG. **72** is a block diagram illustrating a configuration of a divider according to Embodiment 7.

(74) FIG. **73** is a block diagram illustrating a configuration of a geometry information encoder and an attribute information encoder according to Embodiment 7.

(75) FIG. **74** is a block diagram illustrating a configuration of a first decoder according to Embodiment 7.

(76) FIG. **75** is a block diagram illustrating a configuration of a geometry information decoder and an attribute information decoder according to Embodiment 7.

(77) FIG. **76** is a flowchart illustrating an example of a process concerning determination of a quantization value in the encoding of geometry information or the encoding of attribute information according to Embodiment 7.

(78) FIG. **77** is a flowchart illustrating an example of a process of decoding geometry information and attribute information according to Embodiment 7.

(79) FIG. **78** is a diagram for describing a first example of a method of transmitting a quantization parameter according to Embodiment 7.

(80) FIG. **79** is a diagram for describing a second example of the method of transmitting a quantization parameter according to Embodiment 7.

(81) FIG. **80** is a diagram for describing a third example of the method of transmitting a quantization parameter according to Embodiment 7.

(82) FIG. **81** is a flowchart of a process of encoding point cloud data according to Embodiment 7.

(83) FIG. **82** is a flowchart illustrating an example of a process of determining a QP value and updating additional information according to Embodiment 7.

(84) FIG. **83** is a flowchart illustrating an example of a process of encoding according to Embodiment 7.

(85) FIG. **84** is a flowchart illustrating a process of decoding point cloud data according to Embodiment 7.

(86) FIG. **85** is a flowchart illustrating an example of a process of obtaining QP values and

decoding a QP value for a slice or tile according to Embodiment 7.

(87) FIG. **86** is a diagram illustrating a syntax example of GPS according to Embodiment 7.

(88) FIG. **87** is a diagram illustrating a syntax example of APS according to Embodiment 7.

(89) FIG. **88** is a diagram illustrating a syntax example of a header of geometry information according to Embodiment 7.

(90) FIG. **89** is a diagram illustrating a syntax example of a header of attribute information according to Embodiment 7.

(91) FIG. **90** is a diagram for describing another example of the method of transmitting a quantization parameter according to Embodiment 7.

(92) FIG. **91** is a diagram for describing another example of the method of transmitting a quantization parameter according to Embodiment 7.

(93) FIG. **92** is a diagram for describing a ninth example of the method of transmitting a quantization parameter according to Embodiment 7.

(94) FIG. **93** is a diagram for describing an example of control of a QP value according to Embodiment 7.

(95) FIG. **94** is a flowchart illustrating an example of a method of determining a QP value based on the quality of an object according to Embodiment 7.

(96) FIG. **95** is a flowchart illustrating an example of a method of determining a QP value based on a rate control according to Embodiment 7.

(97) FIG. **96** is a flowchart illustrating an encoding process according to Embodiment 7.

(98) FIG. **97** is a flowchart illustrating a decoding process according to Embodiment 7.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(99) A three-dimensional data encoding method according to an aspect of the present disclosure includes: quantizing geometry information of each of three-dimensional points, using a first quantization parameter; quantizing a first luminance using a second quantization parameter and quantizing a first chrominance using a third quantization parameter, the first luminance and the first chrominance indicating a first color among attribute information of each of the three dimensional points; and generating a bitstream including the geometry information quantized, the first luminance quantized, the first chrominance quantized, the first quantization parameter, the second quantization parameter, and a first difference between the second quantization parameter and the third quantization parameter.

(100) According to this three-dimensional data encoding method, since the third quantization parameter is indicated by the first difference from the second quantization parameter in the bitstream, the coding efficiency can be improved.

(101) For example, the three-dimensional data encoding method may further include: quantizing a reflectance among the attribute information of each of the three-dimensional points, using a fourth parameter, wherein in the generating, the bitstream generated may further include the reflectance quantized and the fourth quantization parameter.

(102) For example, in the quantizing using the second quantization parameter, for each of subspaces obtained by dividing a current space including the three-dimensional points, the first luminance of at least one three-dimensional point included in the subspace may be quantized further using a fifth quantization parameter. Furthermore, in the quantizing using the third quantization parameter, the first chrominance of the at least one three-dimensional point may be quantized further using a sixth quantization parameter. Furthermore, in the generating, the bitstream generated may further include a second difference between the second quantization parameter and the fifth quantization parameter and a third difference between the third quantization parameter and the sixth quantization parameter.

(103) According to this three-dimensional data encoding method, since the fifth quantization parameter is indicated by the second difference from the second quantization parameter and the sixth quantization parameter is indicated by the third difference from the third quantization

parameter in the bitstream, the coding efficiency can be improved.

(104) For example, in the generating, the bitstream generated may further include identification information indicating that the quantizing using the second quantization parameter was performed using the fifth quantization parameter and the quantizing using the third quantization parameter was performed using the sixth quantization parameter.

(105) Accordingly, the three-dimensional data decoding device having obtained the bitstream can determine from the identification information that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed, so that the processing load of the decoding process can be reduced.

(106) For example, the three-dimensional data encoding method may further include quantizing a second luminance using a seventh quantization parameter and quantizing a second chrominance using an eighth quantization parameter, the second luminance and the second chrominance indicating a second color among the attribute information of each of the three-dimensional points. Furthermore, in the generating, the bitstream generated may further include the second luminance quantized, the second chrominance quantized, the seventh quantization parameter, and a fourth difference between the seventh parameter and the eighth parameter.

(107) According to this three-dimensional data encoding method, the eighth quantization parameter is indicated by the fourth difference from the seventh quantization parameter in the bitstream, the coding efficiency can be improved. In addition, two types of color information can be included in the attribute information on a three-dimensional point.

(108) Furthermore, a three-dimensional data decoding method according to an aspect of the present disclosure includes: obtaining quantized geometry information, quantized first luminance, quantized first chrominance, a first quantization parameter, a second quantization parameter, and a first difference between the second quantization parameter and a third quantization parameter, by obtaining a bitstream; calculating geometry information of three-dimensional points by inverse-quantizing the quantized geometry information using the first quantization information; calculating a first luminance, out of the first luminance and a first chrominance which indicate a first color of the three-dimensional points, by inverse-quantizing the quantized first luminance using the second quantization parameter; and calculating the first chrominance by inverse-quantizing the quantized first chrominance using the third quantization parameter obtained from the second quantization parameter and the first difference.

(109) In this way, the three-dimensional data decoding method can correctly decode geometry information and attribute information on a three-dimensional point.

(110) For example, in the obtaining, a quantized reflectance and a fourth quantization parameter may be further obtained by obtaining the bitstream, and the three-dimensional data decoding method may further include calculating a reflectance of the three-dimensional points by inverse-quantizing the quantized reflectance using the fourth quantization parameter.

(111) Therefore, the three-dimensional data decoding method can correctly decode the reflectance of a three-dimensional point.

(112) For example, in the obtaining, a second difference between the second quantization parameter and a fifth quantization parameter and a third difference between the third quantization parameter and a sixth quantization parameter may be further obtained by obtaining the bitstream, in the calculating of the first luminance, a first luminance of at least one three-dimensional point may be calculated by inverse-quantizing the quantized first luminance using the fifth quantization parameter obtained from the second quantization parameter and the second difference, the at least one three-dimensional point being included in each of subspaces obtained by dividing a current space including the three-dimensional points, the quantized first luminance being the luminance obtained by quantizing the first luminance of the at least one three-dimensional point using the second quantization parameter and the fifth quantization parameter, and in the calculating of the first chrominance, a first chrominance of the at least one three-dimensional point may be calculated

by inverse-quantizing the quantized first chrominance using the sixth quantization parameter obtained from the third quantization parameter and the third difference, the quantized first chrominance being the chrominance obtained by quantizing the first chrominance of the at least one three-dimensional point using the third quantization parameter and the sixth quantization parameter.

(113) For example, in the obtaining, identification information indicating that quantizing was performed using the fifth quantization parameter and the sixth quantization parameter may be further obtained by obtaining the bitstream. Furthermore, for example, in the calculating of the first luminance, when the identification information indicates that quantizing was performed using the fifth quantization parameter and the sixth quantization parameter, the quantized first luminance may be determined to be the luminance obtained by quantizing the first luminance of the at least one three-dimensional point. Furthermore, for example, in the calculating of the first chrominance, when the identification information indicates that quantizing was performed using the fifth quantization parameter and the sixth quantization parameter, the quantized first chrominance may be determined to be the chrominance obtained by quantizing the first chrominance of the at least one three-dimensional point.

(114) According to this three-dimensional data decoding method, it can be determined from the identification information that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed, so that the processing load of the decoding process can be reduced.

(115) For example, in the obtaining, a quantized second luminance, a quantized second chrominance, a seventh quantization parameter, and a fourth difference between the seventh quantization parameter and an eighth quantization parameter may be further obtained by obtaining the bitstream. Furthermore, for example, the three-dimensional data decoding method may further include: calculating a second luminance out of the second luminance and a second chrominance which indicate a second color of the three-dimensional points, by inverse-quantizing the quantized second luminance using the seventh quantization parameter; and calculating the second chrominance by inverse-quantizing the quantized second chrominance using the eighth quantization parameter obtained from the seventh quantization parameter and the fourth difference.

(116) In this way, the three-dimensional data decoding method can correctly decode the second color of a three-dimensional point.

(117) Furthermore, a three-dimensional data encoding device according to an aspect of the present disclosure includes: a processor; and memory. Here, using the memory, the processor: quantizes geometry information of each of three-dimensional points, using a first quantization parameter; quantizes a first luminance using a second quantization parameter and quantizes a first chrominance using a third quantization parameter, the first luminance and the first chrominance indicating a first color among attribute information of each of the three dimensional points; and generates a bitstream including the geometry information quantized, the first luminance quantized, the first chrominance quantized, the first quantization parameter, the second quantization parameter, and a first difference between the second quantization parameter and the third quantization parameter.

(118) With this three-dimensional data encoding device, since the third quantization parameter is indicated by the first difference from the second quantization parameter, the coding efficiency can be improved.

(119) Furthermore, a three-dimensional data decoding device according to an aspect of the present disclosure includes: a processor; and memory. Here, using the memory, the processor: obtains quantized geometry information, quantized first luminance, quantized first chrominance, a first quantization parameter, a second quantization parameter, and a first difference between the second quantization parameter and a third quantization parameter, by obtaining a bitstream; calculates geometry information of three-dimensional points by inverse-quantizing the quantized geometry

information using the first quantization information; calculates a first luminance, out of the first luminance and a first chrominance which indicate a first color of the three-dimensional points, by inverse-quantizing the quantized first luminance using the second quantization parameter; and calculates the first chrominance by inverse-quantizing the quantized first chrominance using the third quantization parameter obtained from the second quantization parameter and the first difference.

(120) With such a configuration, the three-dimensional data decoding device can correctly decode geometry information and attribute information on a three-dimensional point.

(121) Note that these general or specific aspects may be implemented as a system, a method, an integrated circuit, a computer program, or a computer-readable recording medium such as a CD-ROM, or may be implemented as any combination of a system, a method, an integrated circuit, a computer program, and a recording medium.

(122) The following describes embodiments with reference to the drawings. Note that the following embodiments show exemplary embodiments of the present disclosure. The numerical values, shapes, materials, structural components, the arrangement and connection of the structural components, steps, the processing order of the steps, etc. shown in the following embodiments are mere examples, and thus are not intended to limit the present disclosure. Of the structural components described in the following embodiments, structural components not recited in any one of the independent claims that indicate the broadest concepts will be described as optional structural components.

Embodiment 1

(123) When using encoded data of a point cloud in a device or for a service in practice, required information for the application is desirably transmitted and received in order to reduce the network bandwidth. However, conventional encoding structures for three-dimensional data have no such a function, and there is also no encoding method for such a function.

(124) Embodiment 1 described below relates to a three-dimensional data encoding method and a three-dimensional data encoding device for encoded data of a three-dimensional point cloud that provides a function of transmitting and receiving required information for an application, a three-dimensional data decoding method and a three-dimensional data decoding device for decoding the encoded data, a three-dimensional data multiplexing method for multiplexing the encoded data, and a three-dimensional data transmission method for transmitting the encoded data.

(125) In particular, at present, a first encoding method and a second encoding method are under investigation as encoding methods (encoding schemes) for point cloud data. However, there is no method defined for storing the configuration of encoded data and the encoded data in a system format. Thus, there is a problem that an encoder cannot perform an MUX process (multiplexing), transmission, or accumulation of data.

(126) In addition, there is no method for supporting a format that involves two codecs, the first encoding method and the second encoding method, such as point cloud compression (PCC).

(127) With regard to this embodiment, a configuration of PCC-encoded data that involves two codecs, a first encoding method and a second encoding method, and a method of storing the encoded data in a system format will be described.

(128) A configuration of a three-dimensional data (point cloud data) encoding and decoding system according to this embodiment will be first described. FIG. 1 is a diagram showing an example of a configuration of the three-dimensional data encoding and decoding system according to this embodiment. As shown in FIG. 1, the three-dimensional data encoding and decoding system includes three-dimensional data encoding system **4601**, three-dimensional data decoding system **4602**, sensor terminal **4603**, and external connector **4604**.

(129) Three-dimensional data encoding system **4601** generates encoded data or multiplexed data by encoding point cloud data, which is three-dimensional data. Three-dimensional data encoding system **4601** may be a three-dimensional data encoding device implemented by a single device or a

system implemented by a plurality of devices. The three-dimensional data encoding device may include a part of a plurality of processors included in three-dimensional data encoding system **4601**.

(130) Three-dimensional data encoding system **4601** includes point cloud data generation system **4611**, presenter **4612**, encoder **4613**, multiplexer **4614**, input/output unit **4615**, and controller **4616**. Point cloud data generation system **4611** includes sensor information obtainer **4617**, and point cloud data generator **4618**.

(131) Sensor information obtainer **4617** obtains sensor information from sensor terminal **4603**, and outputs the sensor information to point cloud data generator **4618**. Point cloud data generator **4618** generates point cloud data from the sensor information, and outputs the point cloud data to encoder **4613**.

(132) Presenter **4612** presents the sensor information or point cloud data to a user. For example, presenter **4612** displays information or an image based on the sensor information or point cloud data.

(133) Encoder **4613** encodes (compresses) the point cloud data, and outputs the resulting encoded data, control information (signaling information) obtained in the course of the encoding, and other additional information to multiplexer **4614**. The additional information includes the sensor information, for example.

(134) Multiplexer **4614** generates multiplexed data by multiplexing the encoded data, the control information, and the additional information input thereto from encoder **4613**. A format of the multiplexed data is a file format for accumulation or a packet format for transmission, for example.

(135) Input/output unit **4615** (a communication unit or interface, for example) outputs the multiplexed data to the outside. Alternatively, the multiplexed data may be accumulated in an accumulator, such as an internal memory. Controller **4616** (or an application executor) controls each processor. That is, controller **4616** controls the encoding, the multiplexing, or other processing.

(136) Note that the sensor information may be input to encoder **4613** or multiplexer **4614**. Alternatively, input/output unit **4615** may output the point cloud data or encoded data to the outside as it is.

(137) A transmission signal (multiplexed data) output from three-dimensional data encoding system **4601** is input to three-dimensional data decoding system **4602** via external connector **4604**.

(138) Three-dimensional data decoding system **4602** generates point cloud data, which is three-dimensional data, by decoding the encoded data or multiplexed data. Note that three-dimensional data decoding system **4602** may be a three-dimensional data decoding device implemented by a single device or a system implemented by a plurality of devices. The three-dimensional data decoding device may include a part of a plurality of processors included in three-dimensional data decoding system **4602**.

(139) Three-dimensional data decoding system **4602** includes sensor information obtainer **4621**, input/output unit **4622**, demultiplexer **4623**, decoder **4624**, presenter **4625**, user interface **4626**, and controller **4627**.

(140) Sensor information obtainer **4621** obtains sensor information from sensor terminal **4603**.

(141) Input/output unit **4622** obtains the transmission signal, decodes the transmission signal into the multiplexed data (file format or packet), and outputs the multiplexed data to demultiplexer **4623**.

(142) Demultiplexer **4623** obtains the encoded data, the control information, and the additional information from the multiplexed data, and outputs the encoded data, the control information, and the additional information to decoder **4624**.

(143) Decoder **4624** reconstructs the point cloud data by decoding the encoded data.

(144) Presenter **4625** presents the point cloud data to a user. For example, presenter **4625** displays information or an image based on the point cloud data. User interface **4626** obtains an indication

based on a manipulation by the user. Controller **4627** (or an application executor) controls each processor. That is, controller **4627** controls the demultiplexing, the decoding, the presentation, or other processing.

(145) Note that input/output unit **4622** may obtain the point cloud data or encoded data as it is from the outside. Presenter **4625** may obtain additional information, such as sensor information, and present information based on the additional information. Presenter **4625** may perform a presentation based on an indication from a user obtained on user interface **4626**.

(146) Sensor terminal **4603** generates sensor information, which is information obtained by a sensor. Sensor terminal **4603** is a terminal provided with a sensor or a camera. For example, sensor terminal **4603** is a mobile body, such as an automobile, a flying object, such as an aircraft, a mobile terminal, or a camera.

(147) Sensor information that can be generated by sensor terminal **4603** includes (1) the distance between sensor terminal **4603** and an object or the reflectance of the object obtained by LIDAR, a millimeter wave radar, or an infrared sensor or (2) the distance between a camera and an object or the reflectance of the object obtained by a plurality of monocular camera images or a stereo-camera image, for example. The sensor information may include the posture, orientation, gyro (angular velocity), position (GPS information or altitude), velocity, or acceleration of the sensor, for example. The sensor information may include air temperature, air pressure, air humidity, or magnetism, for example.

(148) External connector **4604** is implemented by an integrated circuit (LSI or IC), an external accumulator, communication with a cloud server via the Internet, or broadcasting, for example.

(149) Next, point cloud data will be described. FIG. 2 is a diagram showing a configuration of point cloud data. FIG. 3 is a diagram showing a configuration example of a data file describing information of the point cloud data.

(150) Point cloud data includes data on a plurality of points. Data on each point includes geometry information (three-dimensional coordinates) and attribute information associated with the geometry information. A set of a plurality of such points is referred to as a point cloud. For example, a point cloud indicates a three-dimensional shape of an object.

(151) Geometry information (position), such as three-dimensional coordinates, may be referred to as geometry. Data on each point may include attribute information (attribute) on a plurality of types of attributes. A type of attribute is color or reflectance, for example.

(152) One piece of attribute information may be associated with one piece of geometry information, or attribute information on a plurality of different types of attributes may be associated with one piece of geometry information. Alternatively, a plurality of pieces of attribute information on the same type of attribute may be associated with one piece of geometry information.

(153) The configuration example of a data file shown in FIG. 3 is an example in which geometry information and attribute information are associated with each other in a one-to-one relationship, and geometry information and attribute information on N points forming point cloud data are shown.

(154) The geometry information is information on three axes, specifically, an x-axis, a y-axis, and a z-axis, for example. The attribute information is RGB color information, for example. A representative data file is ply file, for example.

(155) Next, types of point cloud data will be described. FIG. 4 is a diagram showing types of point cloud data. As shown in FIG. 4, point cloud data includes a static object and a dynamic object.

(156) The static object is three-dimensional point cloud data at an arbitrary time (a time point). The dynamic object is three-dimensional point cloud data that varies with time. In the following, three-dimensional point cloud data associated with a time point will be referred to as a PCC frame or a frame.

(157) The object may be a point cloud whose range is limited to some extent, such as ordinary video data, or may be a large point cloud whose range is not limited, such as map information.

(158) There are point cloud data having varying densities. There may be sparse point cloud data and dense point cloud data.

(159) In the following, each processor will be described in detail. Sensor information is obtained by various means, including a distance sensor such as LIDAR or a range finder, a stereo camera, or a combination of a plurality of monocular cameras. Point cloud data generator **4618** generates point cloud data based on the sensor information obtained by sensor information obtainer **4617**. Point cloud data generator **4618** generates geometry information as point cloud data, and adds attribute information associated with the geometry information to the geometry information.

(160) When generating geometry information or adding attribute information, point cloud data generator **4618** may process the point cloud data. For example, point cloud data generator **4618** may reduce the data amount by omitting a point cloud whose position coincides with the position of another point cloud. Point cloud data generator **4618** may also convert the geometry information (such as shifting, rotating or normalizing the position) or render the attribute information.

(161) Note that, although FIG. 1 shows point cloud data generation system **4611** as being included in three-dimensional data encoding system **4601**, point cloud data generation system **4611** may be independently provided outside three-dimensional data encoding system **4601**.

(162) Encoder **4613** generates encoded data by encoding point cloud data according to an encoding method previously defined. In general, there are the two types of encoding methods described below. One is an encoding method using geometry information, which will be referred to as a first encoding method, hereinafter. The other is an encoding method using a video codec, which will be referred to as a second encoding method, hereinafter.

(163) Decoder **4624** decodes the encoded data into the point cloud data using the encoding method previously defined.

(164) Multiplexer **4614** generates multiplexed data by multiplexing the encoded data in an existing multiplexing method. The generated multiplexed data is transmitted or accumulated. Multiplexer **4614** multiplexes not only the PCC-encoded data but also another medium, such as a video, an audio, subtitles, an application, or a file, or reference time information. Multiplexer **4614** may further multiplex attribute information associated with sensor information or point cloud data.

(165) Multiplexing schemes or file formats include ISOBMFF, MPEG-DASH, which is a transmission scheme based on ISOBMFF, MMT, MPEG-2 TS Systems, or RMP, for example.

(166) Demultiplexer **4623** extracts PCC-encoded data, other media, time information and the like from the multiplexed data.

(167) Input/output unit **4615** transmits the multiplexed data in a method suitable for the transmission medium or accumulation medium, such as broadcasting or communication.

Input/output unit **4615** may communicate with another device over the Internet or communicate with an accumulator, such as a cloud server.

(168) As a communication protocol, http, ftp, TCP, UDP or the like is used. The pull communication scheme or the push communication scheme can be used.

(169) A wired transmission or a wireless transmission can be used. For the wired transmission, Ethernet (registered trademark), USB, RS-232C, HDMI (registered trademark), or a coaxial cable is used, for example. For the wireless transmission, wireless LAN, Wi-Fi (registered trademark), Bluetooth (registered trademark), or a millimeter wave is used, for example.

(170) As a broadcasting scheme, DVB-T2, DVB-S2, DVB-C2, ATSC3.0, or ISDB-S3 is used, for example.

(171) FIG. 5 is a diagram showing a configuration of first encoder **4630**, which is an example of encoder **4613** that performs encoding in the first encoding method. FIG. 6 is a block diagram showing first encoder **4630**. First encoder **4630** generates encoded data (encoded stream) by encoding point cloud data in the first encoding method. First encoder **4630** includes geometry information encoder **4631**, attribute information encoder **4632**, additional information encoder **4633**, and multiplexer **4634**.

(172) First encoder **4630** is characterized by performing encoding by keeping a three-dimensional structure in mind. First encoder **4630** is further characterized in that attribute information encoder **4632** performs encoding using information obtained from geometry information encoder **4631**. The first encoding method is referred to also as geometry-based PCC (GPCC).

(173) Point cloud data is PCC point cloud data like a PLY file or PCC point cloud data generated from sensor information, and includes geometry information (position), attribute information (attribute), and other additional information (metadata). The geometry information is input to geometry information encoder **4631**, the attribute information is input to attribute information encoder **4632**, and the additional information is input to additional information encoder **4633**.

(174) Geometry information encoder **4631** generates encoded geometry information (compressed geometry), which is encoded data, by encoding geometry information. For example, geometry information encoder **4631** encodes geometry information using an N-ary tree structure, such as an octree. Specifically, in the case of an octree, a current space is divided into eight nodes (subspaces), 8-bit information (occupancy code) that indicates whether each node includes a point cloud or not is generated. A node including a point cloud is further divided into eight nodes, and 8-bit information that indicates whether each of the eight nodes includes a point cloud or not is generated. This process is repeated until a predetermined level is reached or the number of the point clouds included in each node becomes equal to or less than a threshold.

(175) Attribute information encoder **4632** generates encoded attribute information (compressed attribute), which is encoded data, by encoding attribute information using configuration information generated by geometry information encoder **4631**. For example, attribute information encoder **4632** determines a reference point (reference node) that is to be referred to in encoding a current point (current node) to be processed based on the octree structure generated by geometry information encoder **4631**. For example, attribute information encoder **4632** refers to a node whose parent node in the octree is the same as the parent node of the current node, of peripheral nodes or neighboring nodes. Note that the method of determining a reference relationship is not limited to this method.

(176) The process of encoding attribute information may include at least one of a quantization process, a prediction process, and an arithmetic encoding process. In this case, “refer to” means using a reference node for calculating a predicted value of attribute information or using a state of a reference node (occupancy information that indicates whether a reference node includes a point cloud or not, for example) for determining a parameter of encoding. For example, the parameter of encoding is a quantization parameter in the quantization process or a context or the like in the arithmetic encoding.

(177) Additional information encoder **4633** generates encoded additional information (compressed metadata), which is encoded data, by encoding compressible data of additional information.

(178) Multiplexer **4634** generates encoded stream (compressed stream), which is encoded data, by multiplexing encoded geometry information, encoded attribute information, encoded additional information, and other additional information. The generated encoded stream is output to a processor in a system layer (not shown).

(179) Next, first decoder **4640**, which is an example of decoder **4624** that performs decoding in the first encoding method, will be described. FIG. 7 is a diagram showing a configuration of first decoder **4640**. FIG. 8 is a block diagram showing first decoder **4640**. First decoder **4640** generates point cloud data by decoding encoded data (encoded stream) encoded in the first encoding method in the first encoding method. First decoder **4640** includes demultiplexer **4641**, geometry information decoder **4642**, attribute information decoder **4643**, and additional information decoder **4644**.

(180) An encoded stream (compressed stream), which is encoded data, is input to first decoder **4640** from a processor in a system layer (not shown).

(181) Demultiplexer **4641** separates encoded geometry information (compressed geometry),

encoded attribute information (compressed attribute), encoded additional information (compressed metadata), and other additional information from the encoded data.

(182) Geometry information decoder **4642** generates geometry information by decoding the encoded geometry information. For example, geometry information decoder **4642** restores the geometry information on a point cloud represented by three-dimensional coordinates from encoded geometry information represented by an N-ary structure, such as an octree.

(183) Attribute information decoder **4643** decodes the encoded attribute information based on configuration information generated by geometry information decoder **4642**. For example, attribute information decoder **4643** determines a reference point (reference node) that is to be referred to in decoding a current point (current node) to be processed based on the octree structure generated by geometry information decoder **4642**. For example, attribute information decoder **4643** refers to a node whose parent node in the octree is the same as the parent node of the current node, of peripheral nodes or neighboring nodes. Note that the method of determining a reference relationship is not limited to this method.

(184) The process of decoding attribute information may include at least one of an inverse quantization process, a prediction process, and an arithmetic decoding process. In this case, “refer to” means using a reference node for calculating a predicted value of attribute information or using a state of a reference node (occupancy information that indicates whether a reference node includes a point cloud or not, for example) for determining a parameter of decoding. For example, the parameter of decoding is a quantization parameter in the inverse quantization process or a context or the like in the arithmetic decoding.

(185) Additional information decoder **4644** generates additional information by decoding the encoded additional information. First decoder **4640** uses additional information required for the decoding process for the geometry information and the attribute information in the decoding, and outputs additional information required for an application to the outside.

(186) Next, second encoder **4650**, which is an example of encoder **4613** that performs encoding in the second encoding method, will be described. FIG. 9 is a diagram showing a configuration of second encoder **4650**. FIG. 10 is a block diagram showing second encoder **4650**.

(187) Second encoder **4650** generates encoded data (encoded stream) by encoding point cloud data in the second encoding method. Second encoder **4650** includes additional information generator **4651**, geometry image generator **4652**, attribute image generator **4653**, video encoder **4654**, additional information encoder **4655**, and multiplexer **4656**.

(188) Second encoder **4650** is characterized by generating a geometry image and an attribute image by projecting a three-dimensional structure onto a two-dimensional image, and encoding the generated geometry image and attribute image in an existing video encoding scheme. The second encoding method is referred to as video-based PCC (VPCC).

(189) Point cloud data is PCC point cloud data like a PLY file or PCC point cloud data generated from sensor information, and includes geometry information (position), attribute information (attribute), and other additional information (metadata).

(190) Additional information generator **4651** generates map information on a plurality of two-dimensional images by projecting a three-dimensional structure onto a two-dimensional image.

(191) Geometry image generator **4652** generates a geometry image based on the geometry information and the map information generated by additional information generator **4651**. The geometry image is a distance image in which distance (depth) is indicated as a pixel value, for example. The distance image may be an image of a plurality of point clouds viewed from one point of view (an image of a plurality of point clouds projected onto one two-dimensional plane), a plurality of images of a plurality of point clouds viewed from a plurality of points of view, or a single image integrating the plurality of images.

(192) Attribute image generator **4653** generates an attribute image based on the attribute information and the map information generated by additional information generator **4651**. The

attribute image is an image in which attribute information (color (RGB), for example) is indicated as a pixel value, for example. The image may be an image of a plurality of point clouds viewed from one point of view (an image of a plurality of point clouds projected onto one two-dimensional plane), a plurality of images of a plurality of point clouds viewed from a plurality of points of view, or a single image integrating the plurality of images.

(193) Video encoder **4654** generates an encoded geometry image (compressed geometry image) and an encoded attribute image (compressed attribute image), which are encoded data, by encoding the geometry image and the attribute image in a video encoding scheme. Note that, as the video encoding scheme, any well-known encoding method can be used. For example, the video encoding scheme is AVC or HEVC.

(194) Additional information encoder **4655** generates encoded additional information (compressed metadata) by encoding the additional information, the map information and the like included in the point cloud data.

(195) Multiplexer **4656** generates an encoded stream (compressed stream), which is encoded data, by multiplexing the encoded geometry image, the encoded attribute image, the encoded additional information, and other additional information. The generated encoded stream is output to a processor in a system layer (not shown).

(196) Next, second decoder **4660**, which is an example of decoder **4624** that performs decoding in the second encoding method, will be described. FIG. **11** is a diagram showing a configuration of second decoder **4660**. FIG. **12** is a block diagram showing second decoder **4660**. Second decoder **4660** generates point cloud data by decoding encoded data (encoded stream) encoded in the second encoding method in the second encoding method. Second decoder **4660** includes demultiplexer **4661**, video decoder **4662**, additional information decoder **4663**, geometry information generator **4664**, and attribute information generator **4665**.

(197) An encoded stream (compressed stream), which is encoded data, is input to second decoder **4660** from a processor in a system layer (not shown).

(198) Demultiplexer **4661** separates an encoded geometry image (compressed geometry image), an encoded attribute image (compressed attribute image), an encoded additional information (compressed metadata), and other additional information from the encoded data.

(199) Video decoder **4662** generates a geometry image and an attribute image by decoding the encoded geometry image and the encoded attribute image in a video encoding scheme. Note that, as the video encoding scheme, any well-known encoding method can be used. For example, the video encoding scheme is AVC or HEVC.

(200) Additional information decoder **4663** generates additional information including map information or the like by decoding the encoded additional information.

(201) Geometry information generator **4664** generates geometry information from the geometry image and the map information. Attribute information generator **4665** generates attribute information from the attribute image and the map information.

(202) Second decoder **4660** uses additional information required for decoding in the decoding, and outputs additional information required for an application to the outside.

(203) In the following, a problem with the PCC encoding scheme will be described. FIG. **13** is a diagram showing a protocol stack relating to PCC-encoded data. FIG. **13** shows an example in which PCC-encoded data is multiplexed with other medium data, such as a video (HEVC, for example) or an audio, and transmitted or accumulated.

(204) A multiplexing scheme and a file format have a function of multiplexing various encoded data and transmitting or accumulating the data. To transmit or accumulate encoded data, the encoded data has to be converted into a format for the multiplexing scheme. For example, with HEVC, a technique for storing encoded data in a data structure referred to as a NAL unit and storing the NAL unit in ISOBMFF is prescribed.

(205) At present, a first encoding method (Coded) and a second encoding method (Codec2) are

under investigation as encoding methods for point cloud data. However, there is no method defined for storing the configuration of encoded data and the encoded data in a system format. Thus, there is a problem that an encoder cannot perform an MUX process (multiplexing), transmission, or accumulation of data.

(206) Note that, in the following, the term “encoding method” means any of the first encoding method and the second encoding method unless a particular encoding method is specified.

(207) In the following, a way of defining a NAL unit according to this embodiment will be described. For example, with a conventional codec, such as HEVC, a NAL unit in one format is defined for one codec. However, there has been no method that supports a format that involves two codecs, that is, the first encoding method and the second encoding method, such as PCC (such a codec will be referred to as a PCC codec, hereinafter).

(208) First, encoder **4670** having the functions of both first encoder **4630** and second encoder **4650** described above and decoder **4680** having the functions of both first decoder **4640** and second decoder **4660** described above will be described.

(209) FIG. **14** is a block diagram showing encoder **4670** according to this embodiment. Encoder **4670** includes first encoder **4630** and second encoder **4650** described above and multiplexer **4671**. Multiplexer **4671** multiplexes encoded data generated by first encoder **4630** and encoded data generated by second encoder **4650**, and outputs the resulting encoded data.

(210) FIG. **15** is a block diagram showing decoder **4680** according to this embodiment. Decoder **4680** includes first decoder **4640** and second decoder **4660** described above and demultiplexer **4681**. Demultiplexer **4681** extracts encoded data generated using the first encoding method and encoded data generated using second encoding method from the input encoded data. Demultiplexer **4681** outputs the encoded data generated using the first encoding method to first decoder **4640**, and outputs the encoded data generated using the second encoding method to second decoder **4660**.

(211) With the configuration described above, encoder **4670** can encode point cloud data by selectively using the first encoding method or the second encoding method. Decoder **4680** can decode encoded data encoded using the first encoding method, encoded data using the second encoding method, and encoded data encoded using both the first encoding method and the second encoding method.

(212) For example, encoder **4670** may change the encoding method (between the first encoding method and the second encoding method) on a point-cloud-data basis or on a frame basis. Alternatively, encoder **4670** may change the encoding method on the basis of an encodable unit.

(213) For example, encoder **4670** generates encoded data (encoded stream) including the identification information for a PCC codec.

(214) Demultiplexer **4681** in decoder **4680** identifies data using the identification information for a PCC codec, for example. When the data is data encoded in the first encoding method, demultiplexer **4681** outputs the data to first decoder **4640**. When the data is data encoded in the second encoding method, demultiplexer **4681** outputs the data to second decoder **4660**.

(215) Encoder **4670** may transmit, as the control information, information indicating whether both the encoding methods are used or any one of the encoding methods is used, in addition to the identification information for the PCC codec.

(216) Next, an encoding process according to this embodiment will be described. FIG. **16** is a flowchart showing an encoding process according to this embodiment. Using the identification information for a PCC codec allows an encoding process ready for a plurality of codecs.

(217) First, encoder **4670** encodes PCC data in both or one of the codecs, that is, the first encoding method and the second encoding method (**S4681**).

(218) When the codec used is the second encoding method (if “second encoding method” in **S4682**), encoder **4670** sets pcc_codec_type in the NAL unit header to a value that indicates that data included in the payload of the NAL unit is data encoded in the second encoding method (**S4683**). Encoder **4670** then sets pcc_nal_unit_type in the NAL unit header to the identifier of the

NAL unit for the second encoding method (S4684). Encoder 4670 then generates a NAL unit having the set NAL unit header and including the encoded data in the payload. Encoder 4670 then transmits the generated NAL unit (S4685).

(219) On the other hand, when the codec used is the first encoding method (if “first encoding method” in S4682), encoder 4670 sets pcc_codec_type in the NAL unit header to a value that indicates that data included in the payload of the NAL unit is data encoded in the first encoding method (S4686). Encoder 4670 then sets pcc_nal_unit_type in the NAL unit header to the identifier of the NAL unit for the first encoding method (S4687). Encoder 4670 then generates a NAL unit having the set NAL unit header and including the encoded data in the payload. Encoder 4670 then transmits the generated NAL unit (S4685).

(220) Next, a decoding process according to this embodiment will be described. FIG. 17 is a flowchart showing a decoding process according to this embodiment. Using the identification information for a PCC codec allows a decoding process ready for a plurality of codecs.

(221) First, decoder 4680 receives a NAL unit (S4691). For example, the NAL unit is the NAL unit generated in the process by encoder 4670 described above.

(222) Decoder 4680 then determines whether pcc_codec_type in the NAL unit header indicates the first encoding method or the second encoding method (S4692).

(223) When pcc_codec_type indicates the second encoding method (if “second encoding method” in S4692), decoder 4680 determines that the data included in the payload of the NAL unit is data encoded in the second encoding method (S4693). Decoder 4680 then identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the second encoding method (S4694). Decoder 4680 then decodes the PCC data in a decoding process for the second encoding method (S4695).

(224) On the other hand, when pcc_codec_type indicates the first encoding method (if “first encoding method” in S4692), decoder 4680 determines that the data included in the payload of the NAL unit is data encoded in the first encoding method (S4696). Decoder 4680 then identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the first encoding method (S4697). Decoder 4680 then decodes the PCC data in a decoding process for the first encoding method (S4698).

(225) As described above, the three-dimensional data encoding device according to an aspect of the present disclosure generates an encoded stream by encoding three-dimensional data (point cloud data, for example), and stores information indicating the encoding method used for the encoding among the first encoding method and the second encoding method (identification information for the codec, for example) in the control information (a parameter set, for example) for the encoded stream.

(226) With such a configuration, the three-dimensional data decoding device can determine the encoding method used for the encoding from the information stored in the control information, when decoding the encoded stream generated by the three-dimensional data encoding device. Therefore, the three-dimensional data decoding device can correctly decode the encoded stream even when a plurality of encoding methods are used.

(227) The three-dimensional data includes geometry information, for example. In the encoding described above, the three-dimensional data encoding device encodes the geometry information. In the storage described above, the three-dimensional data encoding device stores the information indicating the encoding method used for the encoding of the geometry information among the first encoding method and the second encoding method in the control information for the geometry information.

(228) The three-dimensional data includes geometry information and attribute information, for example. In the encoding described above, the three-dimensional data encoding device encodes the geometry information and the attribute information. In the storage described above, the three-dimensional data encoding device stores the information indicating the encoding method used for

the encoding of the geometry information among the first encoding method and the second encoding method in the control information for the geometry information, and stores the information indicating the encoding method used for the encoding of the attribute information among the first encoding method and the second encoding method in the control information for the attribute information.

(229) With such a configuration, different encoding methods can be used for the geometry information and the attribute information, and therefore, the coding efficiency can be improved.

(230) For example, the three-dimensional data encoding method further includes storing the encoded stream in one or more units (NAL units, for example).

(231) For example, the unit includes information (`pcc_nal_unit_type`, for example) indicating the type of data included in the unit that has a format that is common to the first encoding method and the second encoding method and is independently defined for the first encoding method and the second encoding method.

(232) For example, the unit includes information (`codec1_nal_unit_type` or `codec1_nal_unit_type`, for example) indicating the type of data included in the unit that has different formats for the first encoding method and the second encoding method and is independently defined for the first encoding method and the second encoding method.

(233) For example, the unit includes information (`pcc_nal_unit_type`, for example) indicating the type of data included in the unit that has a format that is common to the first encoding method and the second encoding method and is commonly defined for the first encoding method and the second encoding method.

(234) For example, the three-dimensional data encoding device includes a processor and a memory, and the processor performs the processes described above using the memory.

(235) The three-dimensional data decoding device according to this embodiment determines the encoding method used for encoding of an encoded stream obtained by encoding of three-dimensional data based on the information indicating the encoding method used for the encoding of the three-dimensional data among the first encoding method and the second encoding method (identification information for the codec, for example) included in the control information (a parameter set, for example) for the encoded stream, and decodes the encoded stream using the determined encoding method.

(236) With such a configuration, the three-dimensional data decoding device can determine the encoding method used for the encoding from the information stored in the control information, when decoding the encoded stream. Therefore, the three-dimensional data decoding device can correctly decode the encoded stream even when a plurality of encoding methods are used.

(237) The three-dimensional data includes geometry information, and the encoded stream includes encoded data of the geometry information, for example. In the determination described above, the three-dimensional data decoding device determines the encoding method used for the encoding of the geometry information based on the information indicating the encoding method used for the encoding of the geometry information among the first encoding method and the second encoding method included in the control information for the geometry information included in the encoded stream. In the decoding described above, the three-dimensional data decoding device decodes the encoded data of the geometry information using the determined encoding method used for the encoding of the geometry information.

(238) The three-dimensional data includes geometry information and attribute information, and the encoded stream includes encoded data of the geometry information and encoded data of the attribute information, for example. In the determination described above, the three-dimensional data decoding device determines the encoding method used for the encoding of the geometry information based on the information indicating the encoding method used for the encoding of the geometry information among the first encoding method and the second encoding method included in the control information for the geometry information included in the encoded stream, and

determines the encoding method used for the encoding of the attribute information based on the information indicating the encoding method used for the encoding of the attribute information among the first encoding method and the second encoding method included in the control information for the attribute information included in the encoded stream. In the decoding described above, the three-dimensional data decoding device decodes the encoded data of the geometry information using the determined encoding method used for the encoding of the geometry information, and decodes the encoded data of the attribute information using the determined encoding method used for the encoding of the attribute information.

(239) With such a configuration, different encoding methods can be used for the geometry information and the attribute information, and therefore, the coding efficiency can be improved.

(240) For example, the encoded stream is stored in one or more units (NAL units, for example), and the three-dimensional data decoding device further obtains the encoded stream from the one or more units.

(241) For example, the unit includes information (`pcc_nal_unit_type`, for example) indicating the type of data included in the unit that has a format that is common to the first encoding method and the second encoding method and is independently defined for the first encoding method and the second encoding method.

(242) For example, the unit includes information (`codec1_nal_unit_type` or `codec1_nal_unit_type`, for example) indicating the type of data included in the unit that has different formats for the first encoding method and the second encoding method and is independently defined for the first encoding method and the second encoding method.

(243) For example, the unit includes information (`pcc_nal_unit_type`, for example) indicating the type of data included in the unit that has a format that is common to the first encoding method and the second encoding method and is commonly defined for the first encoding method and the second encoding method.

(244) For example, the three-dimensional data decoding device includes a processor and a memory, and the processor performs the processes described above using the memory.

Embodiment 2

(245) In Embodiment 2, a method of storing the NAL unit in an ISOBMFF file will be described.

(246) ISOBMFF is a file format standard prescribed in ISO/IEC14496-12. ISOBMFF is a standard that does not depend on any medium, and prescribes a format that allows various media, such as a video, an audio, and a text, to be multiplexed and stored.

(247) A basic structure (file) of ISOBMFF will be described. A basic unit of ISOBMFF is a box. A box is formed by type, length, and data, and a file is a set of various types of boxes.

(248) FIG. **18** is a diagram showing a basic structure (file) of ISOBMFF. A file in ISOBMFF includes boxes, such as `ftyp` that indicates the brand of the file by four-character code (4CC), `moov` that stores metadata, such as control information (signaling information), and `mdat` that stores data.

(249) A method for storing each medium in the ISOBMFF file is separately prescribed. For example, a method of storing an AVC video or an HEVC video is prescribed in ISO/IEC14496-15. Here, it can be contemplated to expand the functionality of ISOBMFF and use ISOBMFF to accumulate or transmit PCC-encoded data. However, there has been no convention for storing PCC-encoded data in an ISOBMFF file. In this embodiment, a method of storing PCC-encoded data in an ISOBMFF file will be described.

(250) FIG. **19** is a diagram showing a protocol stack in a case where a common PCC codec NAL unit in an ISOBMFF file. Here, a common PCC codec NAL unit is stored in an ISOBMFF file. Although the NAL unit is common to PCC codecs, a storage method for each codec (Carriage of Codec1, Carriage of Codec2) is desirably prescribed, since a plurality of PCC codecs are stored in the NAL unit.

(251) Next, a method of storing a common PCC NAL unit that supports a plurality of PCC codecs in an ISOBMFF file will be described. FIG. **20** is a diagram showing an example in which a

common PCC NAL unit is stored in an ISOBMFF file for the storage method for codec 1 (Carriage of Coded). FIG. 21 is a diagram showing an example in which a common PCC NAL unit is stored in an ISOBMFF file for the storage method for codec 2 (Carriage of Codec2).

(252) Here, ftyp is information that is important for identification of the file format, and a different identifier of ftyp is defined for each codec. When PCC-encoded data encoded in the first encoding method (encoding scheme) is stored in the file, ftyp is set to pcc1. When PCC-encoded data encoded in the second encoding method is stored in the file, ftyp is set to pcc2.

(253) Here, pcc1 indicates that PCC codec 1 (first encoding method) is used. pcc2 indicates that PCC codec2 (second encoding method) is used. That is, pcc1 and pcc2 indicate that the data is PCC (encoded three-dimensional data (point cloud data)), and indicate the PCC codec (first encoding method or second encoding method).

(254) In the following, a method of storing a NAL unit in an ISOBMFF file will be described. The multiplexer analyzes the NAL unit header, and describes pcc1 in ftyp of ISOBMFF if pcc_codec_type=Codec1.

(255) The multiplexer analyzes the NAL unit header, and describes pcc2 in ftyp of ISOBMFF if pcc_codec_type=Codec2.

(256) If pcc_nal_unit_type is metadata, the multiplexer stores the NAL unit in moov or mdat in a predetermined manner, for example. If pcc_nal_unit_type is data, the multiplexer stores the NAL unit in moov or mdat in a predetermined manner, for example.

(257) For example, the multiplexer may store the NAL unit size in the NAL unit, as with HEVC.

(258) According to this storage method, the demultiplexer (a system layer) can determine whether the PCC-encoded data is encoded in the first encoding method or the second encoding method by analyzing ftyp included in the file. Furthermore, as described above, by determining whether the PCC-encoded data is encoded in the first encoding method or the second encoding method, the encoded data encoded in any one of the encoding methods can be extracted from the data including both the encoded data encoded in the encoding methods. Therefore, when transmitting the encoded data, the amount of data transmitted can be reduced. In addition, according to this storage method, different data (file) formats do not need to be set for the first encoding method and the second encoding method, and a common data format can be used for the first encoding method and the second encoding method.

(259) Note that, when the identification information for the codec, such as ftyp of ISOBMFF, is indicated in the metadata of the system layer, the multiplexer can store a NAL unit without pcc_nal_unit_type in the ISOBMFF file.

(260) Next, configurations and operations of the multiplexer of the three-dimensional data encoding system (three-dimensional data encoding device) according to this embodiment and the demultiplexer of the three-dimensional data decoding system (three-dimensional data decoding device) according to this embodiment will be described.

(261) FIG. 22 is a diagram showing a configuration of first multiplexer **4710**. First multiplexer **4710** includes file converter **4711** that generates multiplexed data (file) by storing encoded data generated by first encoder **4630** and control information (NAL unit) in an ISOBMFF file. First multiplexer **4710** is included in multiplexer **4614** shown in FIG. 1, for example.

(262) FIG. 23 is a diagram showing a configuration of first demultiplexer **4720**. First demultiplexer **4720** includes file inverse converter **4721** that obtains encoded data and control information (NAL unit) from multiplexed data (file) and outputs the obtained encoded data and control information to first decoder **4640**. First demultiplexer **4720** is included in demultiplexer **4623** shown in FIG. 1, for example.

(263) FIG. 24 is a diagram showing a configuration of second multiplexer **4730**. Second multiplexer **4730** includes file converter **4731** that generates multiplexed data (file) by storing encoded data generated by second encoder **4650** and control information (NAL unit) in an ISOBMFF file. Second multiplexer **4730** is included in multiplexer **4614** shown in FIG. 1, for

example.

(264) FIG. 25 is a diagram showing a configuration of second demultiplexer **4740**. Second demultiplexer **4740** includes file inverse converter **4741** that obtains encoded data and control information (NAL unit) from multiplexed data (file) and outputs the obtained encoded data and control information to second decoder **4660**. Second demultiplexer **4740** is included in demultiplexer **4623** shown in FIG. 1, for example.

(265) FIG. 26 is a flowchart showing a multiplexing process by first multiplexer **4710**. First, first multiplexer **4710** analyzes pcc_codec_type in the NAL unit header, thereby determining whether the codec used is the first encoding method or the second encoding method (S4701).

(266) When pcc_codec_type represents the second encoding method (if “second encoding method” in S4702), first multiplexer **4710** does not process the NAL unit (S4703).

(267) On the other hand, when pcc_codec_type represents the first encoding method (if “first encoding method” in S4702), first multiplexer **4710** describes pcc1 in ftyp (S4704). That is, first multiplexer **4710** describes information indicating that data encoded in the first encoding method is stored in the file in ftyp.

(268) First multiplexer **4710** then analyzes pcc_nal_unit_type in the NAL unit header, and stores the data in a box (moov or mdat, for example) in a predetermined manner suitable for the data type represented by pcc_nal_unit_type (S4705). First multiplexer **4710** then creates an ISOBMFF file including the ftyp described above and the box described above (S4706).

(269) FIG. 27 is a flowchart showing a multiplexing process by second multiplexer **4730**. First, second multiplexer **4730** analyzes pcc_codec_type in the NAL unit header, thereby determining whether the codec used is the first encoding method or the second encoding method (S4711).

(270) When pcc_codec_type represents the second encoding method (if “second encoding method” in S4712), second multiplexer **4730** describes pcc2 in ftyp (S4713). That is, second multiplexer **4730** describes information indicating that data encoded in the second encoding method is stored in the file in ftyp.

(271) Second multiplexer **4730** then analyzes pcc_nal_unit_type in the NAL unit header, and stores the data in a box (moov or mdat, for example) in a predetermined manner suitable for the data type represented by pcc_nal_unit_type (S4714). Second multiplexer **4730** then creates an ISOBMFF file including the ftyp described above and the box described above (S4715).

(272) On the other hand, when pcc_codec_type represents the first encoding method (if “first encoding method” in S4712), second multiplexer **4730** does not process the NAL unit (S4716).

(273) Note that the process described above is an example in which PCC data is encoded in any one of the first encoding method and the second encoding method. First multiplexer **4710** and second multiplexer **4730** store a desired NAL unit in a file by identifying the codec type of the NAL unit. Note that, when the identification information for the PCC codec is included in a location other than the NAL unit header, first multiplexer **4710** and second multiplexer **4730** may identify the codec type (first encoding method or second encoding method) based on the identification information for the PCC codec included in the location other than the NAL unit header in step S4701 or S4711.

(274) When storing data in a file in step S4706 or S4714, first multiplexer **4710** and second multiplexer **4730** may store the data in the file after deleting pcc_nal_unit_type from the NAL unit header.

(275) FIG. 28 is a flowchart showing a process performed by first demultiplexer **4720** and first decoder **4640**. First, first demultiplexer **4720** analyzes ftyp in an ISOBMFF file (S4721). When the codec represented by ftyp is the second encoding method (pcc2) (if “second encoding method” in S4722), first demultiplexer **4720** determines that the data included in the payload of the NAL unit is data encoded in the second encoding method (S4723). First demultiplexer **4720** also transmits the result of the determination to first decoder **4640**. First decoder **4640** does not process the NAL unit (S4724).

(276) On the other hand, when the codec represented by ftyp is the first encoding method (pcc1) (if “first encoding method” in S4722), first demultiplexer 4720 determines that the data included in the payload of the NAL unit is data encoded in the first encoding method (S4725). First demultiplexer 4720 also transmits the result of the determination to first decoder 4640.

(277) First decoder 4640 identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the first encoding method (S4726). First decoder 4640 then decodes the PCC data using a decoding process for the first encoding method (S4727).

(278) FIG. 29 is a flowchart showing a process performed by second demultiplexer 4740 and second decoder 4660. First, second demultiplexer 4740 analyzes ftyp in an ISOBMFF file (S4731). When the codec represented by ftyp is the second encoding method (pcc2) (if “second encoding method” in S4732), second demultiplexer 4740 determines that the data included in the payload of the NAL unit is data encoded in the second encoding method (S4733). Second demultiplexer 4740 also transmits the result of the determination to second decoder 4660.

(279) Second decoder 4660 identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the second encoding method (S4734). Second decoder 4660 then decodes the PCC data using a decoding process for the second encoding method (S4735).

(280) On the other hand, when the codec represented by ftyp is the first encoding method (pcc1) (if “first encoding method” in S4732), second demultiplexer 4740 determines that the data included in the payload of the NAL unit is data encoded in the first encoding method (S4736). Second demultiplexer 4740 also transmits the result of the determination to second decoder 4660. Second decoder 4660 does not process the NAL unit (S4737).

(281) As described above, for example, since the codec type of the NAL unit is identified in first demultiplexer 4720 or second demultiplexer 4740, the codec type can be identified in an early stage. Furthermore, a desired NAL unit can be input to first decoder 4640 or second decoder 4660, and an unwanted NAL unit can be removed. In this case, the process of first decoder 4640 or second decoder 4660 analyzing the identification information for the codec may be unnecessary. Note that a process of referring to the NAL unit type again and analyzing the identification information for the codec may be performed by first decoder 4640 or second decoder 4660.

(282) Furthermore, if pcc_nal_unit_type is deleted from the NAL unit header by first multiplexer 4710 or second multiplexer 4730, first demultiplexer 4720 or second demultiplexer 4740 can output the NAL unit to first decoder 4640 or second decoder 4660 after adding pcc_nal_unit_type to the NAL unit.

Embodiment 3

(283) In Embodiment 3, a multiplexer and a demultiplexer that correspond to encoder 4670 and decoder 4680 ready for a plurality of codecs described above with regard to Embodiment 1 will be described. FIG. 30 is a diagram showing configurations of encoder 4670 and third multiplexer 4750 according to this embodiment.

(284) Encoder 4670 encodes point cloud data in both or one of the first encoding method and the second encoding method. Encoder 4670 may change the encoding method (between the first encoding method and the second encoding method) on a point-cloud-data basis or on a frame basis. Alternatively, encoder 4670 may change the encoding method on the basis of an encodable unit.

(285) Encoder 4670 generates encoded data (encoded stream) including the identification information for a PCC codec.

(286) Third multiplexer 4750 includes file converter 4751. File converter 4751 converts a NAL unit output from encoder 4670 into a PCC data file. File converter 4751 analyzes the codec identification information included in the NAL unit header, and determines whether the PCC-encoded data is data encoded in the first encoding method, data encoded in the second encoding method, or data encoded in both the encoding methods. File converter 4751 describes a brand name

that allows codec identification in ftyp. For example, when indicating the data is encoded in both the encoding methods, pcc3 is described in ftyp.

(287) Note that, when encoder **4670** describes the PCC codec identification information in a location other than the NAL unit, file converter **4751** may determine the PCC codec (encoding method) based on the identification information.

(288) FIG. **31** is a diagram showing configurations of third demultiplexer **4760** and decoder **4680** according to this embodiment.

(289) Third demultiplexer **4760** includes file inverse converter **4761**. File inverse converter **4761** analyzes ftyp included in a file, and determines whether the PCC-encoded data is data encoded in the first encoding method, data encoded in the second encoding method, or data encoded in both the encoding methods.

(290) When the PCC-encoded data is data encoded in any one of the encoding methods, the data is input to an appropriate one of first decoder **4640** and second decoder **4660**, and is not input to the other decoder. When the PCC-encoded data is data encoded in both the encoding methods, the data is input to decoder **4680** ready for both the encoding methods.

(291) Decoder **4680** decodes the PCC-encoded data in both or one of the first encoding method and the second encoding method.

(292) FIG. **32** is a flowchart showing a process performed by third multiplexer **4750** according to this embodiment.

(293) First, third multiplexer **4750** analyzes pcc_codec_type in the NAL unit header, thereby determining whether the codec(s) used is the first encoding method, the second encoding method, or both the first encoding method and the second encoding method (**S4741**).

(294) When the second encoding method is used (if Yes in **S4742** and “second encoding method” in **S4743**), third multiplexer **4750** describes pcc2 in ftyp (**S4744**). That is, third multiplexer **4750** describes information indicating that data encoded in the second encoding method is stored in the file in ftyp.

(295) Third multiplexer **4750** then analyzes pcc_nal_unit_type in the NAL unit header, and stores the data in a box (moov or mdat, for example) in a predetermined manner suitable for the data type represented by pcc_nal_unit_type (**S4745**). Third multiplexer **4750** then creates an ISOBMFF file including the ftyp described above and the box described above (**S4746**).

(296) When the first encoding method is used (if Yes in **S4742** and “first encoding method” in **S4743**), third multiplexer **4750** describes pcc1 in ftyp (**S4747**). That is, third multiplexer **4750** describes information indicating that data encoded in the first encoding method is stored in the file in ftyp.

(297) Third multiplexer **4750** then analyzes pcc_nal_unit_type in the NAL unit header, and stores the data in a box (moov or mdat, for example) in a predetermined manner suitable for the data type represented by pcc_nal_unit_type (**S4748**). Third multiplexer **4750** then creates an ISOBMFF file including the ftyp described above and the box described above (**S4746**).

(298) When both the first encoding method and the second encoding method are used (if No in **S4742**), third multiplexer **4750** describes pcc3 in ftyp (**S4749**). That is, third multiplexer **4750** describes information indicating that data encoded in both the encoding methods is stored in the file in ftyp.

(299) Third multiplexer **4750** then analyzes pcc_nal_unit_type in the NAL unit header, and stores the data in a box (moov or mdat, for example) in a predetermined manner suitable for the data type represented by pcc_nal_unit_type (**S4750**). Third multiplexer **4750** then creates an ISOBMFF file including the ftyp described above and the box described above (**S4746**).

(300) FIG. **33** is a flowchart showing a process performed by third demultiplexer **4760** and decoder **4680**. First, third demultiplexer **4760** analyzes ftyp included in an ISOBMFF file (**S4761**). When the codec represented by ftyp is the second encoding method (pcc2) (if Yes in **S4762** and “second encoding method” in **S4763**), third demultiplexer **4760** determines that the data included in the

payload of the NAL unit is data encoded in the second encoding method (S4764). Third demultiplexer **4760** also transmits the result of the determination to decoder **4680**.

(301) Decoder **4680** identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the second encoding method (S4765). Decoder **4680** then decodes the PCC data using a decoding process for the second encoding method (S4766).

(302) When the codec represented by ftyp is the first encoding method (pcc1) (if Yes in S4762 and “first encoding method” in S4763), third demultiplexer **4760** determines that the data included in the payload of the NAL unit is data encoded in the first encoding method (S4767). Third demultiplexer **4760** also transmits the result of the determination to decoder **4680**.

(303) Decoder **4680** identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the first encoding method (S4768). Decoder **4680** then decodes the PCC data using a decoding process for the first encoding method (S4769).

(304) When ftyp indicates that both the encoding methods are used (pcc3) (if No in S4762), third demultiplexer **4760** determines that the data included in the payload of the NAL unit is data encoded in both the first encoding method and the second encoding method (S4770). Third demultiplexer **4760** also transmits the result of the determination to decoder **4680**.

(305) Decoder **4680** identifies the data based on the determination that pcc_nal_unit_type in the NAL unit header is the identifier of the NAL unit for the codecs described in pcc_codec_type (S4771). Decoder **4680** then decodes the PCC data using decoding processes for both the encoding methods (S4772). That is, decoder **4680** decodes the data encoded in the first encoding method using a decoding process for the first encoding method, and decodes the data encoded in the second encoding method using a decoding process for the second encoding method.

(306) In the following, variations of this embodiment will be described. As types of brands represented by ftyp, the types described below can be indicated by the identification information. Furthermore, a combination of a plurality of the types described below can also be indicated by the identification information.

(307) The identification information may indicate whether the original data object yet to be PCC-encoded is a point cloud whose range is limited or a large point cloud whose range is not limited, such as map information.

(308) The identification information may indicate whether the original data yet to be PCC-encoded is a static object or a dynamic object.

(309) As described above, the identification information may indicate whether the PCC-encoded data is data encoded in the first encoding method or data encoded in the second encoding method.

(310) The identification information may indicate an algorithm used in the PCC encoding. Here, the “algorithm” means an encoding method that can be used in the first encoding method or the second encoding method, for example.

(311) The identification information may indicate a differentiation between methods of storing the PCC-encoded data into an ISO BMFF file. For example, the identification information may indicate whether the storage method used is a storage method for accumulation or a storage method for real-time transmission, such as dynamic streaming.

(312) Although an example in which ISO BMFF is used as a file format has been described in Embodiments 2 and 3, other formats can also be used. For example, the method according to this embodiment can also be used when PCC-encoded data is stored in MPEG-2 TS Systems, MPEG-DASH, MMT, or RMP.

(313) Although an example in which metadata, such as the identification information, is stored in ftyp has been shown above, metadata can also be stored in a location other than ftyp. For example, the metadata may be stored in moov.

(314) As described above, a three-dimensional data storing device (or three-dimensional data multiplexing device or three-dimensional data encoding device) performs the process shown in

FIG. 34.

(315) First, the three-dimensional data storing device (which includes first multiplexer **4710**, second multiplexer **4730** or third multiplexer **4750**, for example) acquires one or more units (NAL units, for example) that store an encoded stream, which is encoded point cloud data (**S4781**). The three-dimensional data storing device then stores the one or more units in a file (an ISOBMFF file, for example) (**S4782**). In the storage (**S4782**), the three-dimensional data storing device also stores information indicating that the data stored in the file is encoded point cloud data (pcc1, pcc2, or pcc3, for example) in the control information (ftyp, for example) (referred to also as signaling information) for the file.

(316) With such a configuration, a device that processes the file generated by the three-dimensional data storing device can quickly determine whether the data stored in the file is encoded point cloud data or not by referring to the control information for the file. Therefore, the processing amount of the device can be reduced, or the processing speed of the device can be increased.

(317) For example, the information indicates the encoding method used for the encoding of the point cloud data among the first encoding method and the second encoding method. Note that the fact that the data stored in the file is encoded point cloud data and the encoding method used for the encoding of the point cloud data among the first encoding method and the second encoding method may be indicated by a single piece of information or different pieces of information.

(318) With such a configuration, a device that processes the file generated by the three-dimensional data storing device can quickly determine the codec used for the data stored in the file by referring to the control information for the file. Therefore, the processing amount of the device can be reduced, or the processing speed of the device can be increased.

(319) For example, the first encoding method is a method (GPCC) that encodes geometry information that represents the position of point cloud data as an N-ary tree (N represents an integer equal to or greater than 2) and encodes attribute information using the geometry information, and the second encoding method is a method (VPCC) that generates a two-dimensional image from point cloud data and encodes the two-dimensional image in a video encoding method.

(320) For example, the file described above is in conformity with ISOBMFF (ISO-based media file format).

(321) For example, the three-dimensional data storing device includes a processor and a memory, and the processor performs the processes described above using the memory.

(322) As described above, a three-dimensional data acquisition device (or three-dimensional data demultiplexing device or three-dimensional data decoding device) performs the process shown in FIG. 35.

(323) The three-dimensional data acquisition device (which includes first demultiplexer **4720**, second demultiplexer **4740**, or third demultiplexer **4760**, for example) acquires a file (an ISOBMFF file, for example) that stores one or more units (NAL units, for example) that store an encoded stream, which is encoded point cloud data (**S4791**). The three-dimensional data acquisition device acquires the one or more units from the file (**S4792**). The control information (ftyp, for example) for the file includes information indicating that the data stored in the file is encoded point cloud data (pcc1, pcc2, or pcc3, for example).

(324) For example, the three-dimensional data acquisition device determines whether the data stored in the file is encoded point cloud data or not by referring to the information. When the three-dimensional data acquisition device determines that the data stored in the file is encoded point cloud data, the three-dimensional data acquisition device generates point cloud data by decoding the encoded point cloud data included in the one or more units. Alternatively, when the three-dimensional data acquisition device determines that the data stored in the file is encoded point cloud data, the three-dimensional data acquisition device outputs information indicating that the data included in the one or more units is encoded point cloud data to a processor in a subsequent stage (first decoder **4640**, second decoder **4660**, or decoder **4680**, for example) (or notifies a

processor in a subsequent stage that the data included in the one or more units is encoded point cloud data).

(325) With such a configuration, the three-dimensional data acquisition device can quickly determine whether the data stored in the file is encoded point cloud data or not by referring to the control information for the file. Therefore, the processing amount of the three-dimensional data acquisition device or a device in a subsequent stage can be reduced, or the processing speed of the three-dimensional data acquisition device or a device in a subsequent stage can be increased.

(326) For example, the information represents the encoding method used for the encoding among the first encoding method and the second encoding method. Note that the fact that the data stored in the file is encoded point cloud data and the encoding method used for the encoding of the point cloud data among the first encoding method and the second encoding method may be indicated by a single piece of information or different pieces of information.

(327) With such a configuration, the three-dimensional data acquisition device can quickly determine the codec used for the data stored in the file by referring to the control information for the file. Therefore, the processing amount of the three-dimensional data acquisition device or a device in a subsequent stage can be reduced, or the processing speed of the three-dimensional data acquisition device or a device in a subsequent stage can be increased.

(328) For example, based on the information, the three-dimensional data acquisition device acquires the data encoded in any one of the first encoding method and the second encoding method from the encoded point cloud data including the data encoded in the first encoding method and the data encoded in the second encoding method.

(329) For example, the first encoding method is a method (GPCC) that encodes geometry information that represents the position of point cloud data as an N-ary tree (N represents an integer equal to or greater than 2) and encodes attribute information using the geometry information, and the second encoding method is a method (VPCC) that generates a two-dimensional image from point cloud data and encodes the two-dimensional image in a video encoding method.

(330) For example, the file described above is in conformity with ISOBMFF (ISO-based media file format).

(331) For example, the three-dimensional data acquisition device includes a processor and a memory, and the processor performs the processes described above using the memory.

Embodiment 4

(332) In Embodiment 4, types of the encoded data (geometry information (geometry), attribute information (attribute), and additional information (metadata)) generated by first encoder **4630** or second encoder **4650** described above, a method of generating additional information (metadata), and a multiplexing process in the multiplexer will be described. The additional information (metadata) may be referred to as a parameter set or control information (signaling information).

(333) In this embodiment, the dynamic object (three-dimensional point cloud data that varies with time) described above with reference to FIG. 4 will be described, for example. However, the same method can also be used for the static object (three-dimensional point cloud data associated with an arbitrary time point).

(334) FIG. 36 is a diagram showing configurations of encoder **4801** and multiplexer **4802** in a three-dimensional data encoding device according to this embodiment. Encoder **4801** corresponds to first encoder **4630** or second encoder **4650** described above, for example. Multiplexer **4802** corresponds to multiplexer **4634** or **4656** described above.

(335) Encoder **4801** encodes a plurality of PCC (point cloud compression) frames of point cloud data to generate a plurality of pieces of encoded data (multiple compressed data) of geometry information, attribute information, and additional information.

(336) Multiplexer **4802** integrates a plurality of types of data (geometry information, attribute information, and additional information) into a NAL unit, thereby converting the data into a data configuration that takes data access in the decoding device into consideration.

(337) FIG. 37 is a diagram showing a configuration example of the encoded data generated by encoder **4801**. Arrows in the drawing indicate a dependence involved in decoding of the encoded data. The source of an arrow depends on data of the destination of the arrow. That is, the decoding device decodes the data of the destination of an arrow, and decodes the data of the source of the arrow using the decoded data. In other words, “a first entity depends on a second entity” means that data of the second entity is referred to (used) in processing (encoding, decoding, or the like) of data of the first entity.

(338) First, a process of generating encoded data of geometry information will be described. Encoder **4801** encodes geometry information of each frame to generate encoded geometry data (compressed geometry data) for each frame. The encoded geometry data is denoted by $G(i)$. i denotes a frame number or a time point of a frame, for example.

(339) Furthermore, encoder **4801** generates a geometry parameter set ($GPS(i)$) for each frame. The geometry parameter set includes a parameter that can be used for decoding of the encoded geometry data. The encoded geometry data for each frame depends on an associated geometry parameter set.

(340) The encoded geometry data formed by a plurality of frames is defined as a geometry sequence. Encoder **4801** generates a geometry sequence parameter set (referred to also as geometry sequence PS or geometry SPS) that stores a parameter commonly used for a decoding process for the plurality of frames in the geometry sequence. The geometry sequence depends on the geometry SPS.

(341) Next, a process of generating encoded data of attribute information will be described. Encoder **4801** encodes attribute information of each frame to generate encoded attribute data (compressed attribute data) for each frame. The encoded attribute data is denoted by $A(i)$. FIG. 37 shows an example in which there are attribute X and attribute Y, and encoded attribute data for attribute X is denoted by $AX(i)$, and encoded attribute data for attribute Y is denoted by $AY(i)$.

(342) Furthermore, encoder **4801** generates an attribute parameter set ($APS(i)$) for each frame. The attribute parameter set for attribute X is denoted by $AXPS(i)$, and the attribute parameter set for attribute Y is denoted by $AYPS(i)$. The attribute parameter set includes a parameter that can be used for decoding of the encoded attribute information. The encoded attribute data depends on an associated attribute parameter set.

(343) The encoded attribute data formed by a plurality of frames is defined as an attribute sequence. Encoder **4801** generates an attribute sequence parameter set (referred to also as attribute sequence PS or attribute SPS) that stores a parameter commonly used for a decoding process for the plurality of frames in the attribute sequence. The attribute sequence depends on the attribute SPS.

(344) In the first encoding method, the encoded attribute data depends on the encoded geometry data.

(345) FIG. 37 shows an example in which there are two types of attribute information (attribute X and attribute Y). When there are two types of attribute information, for example, two encoders generate data and metadata for the two types of attribute information. For example, an attribute sequence is defined for each type of attribute information, and an attribute SPS is generated for each type of attribute information.

(346) Note that, although FIG. 37 shows an example in which there is one type of geometry information, and there are two types of attribute information, the present invention is not limited thereto. There may be one type of attribute information or three or more types of attribute information. In such cases, encoded data can be generated in the same manner. If the point cloud data has no attribute information, there may be no attribute information. In such a case, encoder **4801** does not have to generate a parameter set associated with attribute information.

(347) Next, a process of generating encoded data of additional information (metadata) will be described. Encoder **4801** generates a PCC stream PS (referred to also as PCC stream PS or stream

PS), which is a parameter set for the entire PCC stream. Encoder **4801** stores a parameter that can be commonly used for a decoding process for one or more geometry sequences and one or more attribute sequences in the stream PS. For example, the stream PS includes identification information indicating the codec for the point cloud data and information indicating an algorithm used for the encoding, for example. The geometry sequence and the attribute sequence depend on the stream PS.

(348) Next, an access unit and a GOF will be described. In this embodiment, concepts of access unit (AU) and group of frames (GOF) are newly introduced.

(349) An access unit is a basic unit for accessing data in decoding, and is formed by one or more pieces of data and one or more pieces of metadata. For example, an access unit is formed by geometry information and one or more pieces of attribute information associated with a same time point. A GOF is a random access unit, and is formed by one or more access units.

(350) Encoder **4801** generates an access unit header (AU header) as identification information indicating the top of an access unit. Encoder **4801** stores a parameter relating to the access unit in the access unit header. For example, the access unit header includes a configuration of or information on the encoded data included in the access unit. The access unit header further includes a parameter commonly used for the data included in the access unit, such as a parameter relating to decoding of the encoded data.

(351) Note that encoder **4801** may generate an access unit delimiter that includes no parameter relating to the access unit, instead of the access unit header. The access unit delimiter is used as identification information indicating the top of the access unit. The decoding device identifies the top of the access unit by detecting the access unit header or the access unit delimiter.

(352) Next, generation of identification information for the top of a GOF will be described. As identification information indicating the top of a GOF, encoder **4801** generates a GOF header. Encoder **4801** stores a parameter relating to the GOF in the GOF header. For example, the GOF header includes a configuration of or information on the encoded data included in the GOF. The GOF header further includes a parameter commonly used for the data included in the GOF, such as a parameter relating to decoding of the encoded data.

(353) Note that encoder **4801** may generate a GOF delimiter that includes no parameter relating to the GOF, instead of the GOF header. The GOF delimiter is used as identification information indicating the top of the GOF. The decoding device identifies the top of the GOF by detecting the GOF header or the GOF delimiter.

(354) In the PCC-encoded data, the access unit is defined as a PCC frame unit, for example. The decoding device accesses a PCC frame based on the identification information for the top of the access unit.

(355) For example, the GOF is defined as one random access unit. The decoding device accesses a random access unit based on the identification information for the top of the GOF. For example, if PCC frames are independent from each other and can be separately decoded, a PCC frame can be defined as a random access unit.

(356) Note that two or more PCC frames may be assigned to one access unit, and a plurality of random access units may be assigned to one GOF.

(357) Encoder **4801** may define and generate a parameter set or metadata other than those described above. For example, encoder **4801** may generate supplemental enhancement information (SEI) that stores a parameter (an optional parameter) that is not always used for decoding.

(358) Next, a configuration of encoded data and a method of storing encoded data in a NAL unit will be described.

(359) For example, a data format is defined for each type of encoded data. FIG. **38** is a diagram showing an example of encoded data and a NAL unit.

(360) For example, as shown in FIG. **38**, encoded data includes a header and a payload. The encoded data may include length information indicating the length (data amount) of the encoded

data, the header, or the payload. The encoded data may include no header.

(361) The header includes identification information for identifying the data, for example. The identification information indicates a data type or a frame number, for example.

(362) The header includes identification information indicating a reference relationship, for example. The identification information is stored in the header when there is a dependence relationship between data, for example, and allows an entity to refer to another entity. For example, the header of the entity to be referred to includes identification information for identifying the data. The header of the referring entity includes identification information indicating the entity to be referred to.

(363) Note that, when the entity to be referred to or the referring entity can be identified or determined from other information, the identification information for identifying the data or identification information indicating the reference relationship can be omitted.

(364) Multiplexer **4802** stores the encoded data in the payload of the NAL unit. The NAL unit header includes `pcc_nal_unit_type`, which is identification information for the encoded data. FIG. **39** is a diagram showing a semantics example of `pcc_nal_unit_type`.

(365) As shown in FIG. **39**, when `pcc_codec_type` is codec 1 (Codec1: first encoding method), values 0 to 10 of `pcc_nal_unit_type` are assigned to encoded geometry data (Geometry), encoded attribute X data (AttributeX), encoded attribute Y data (AttributeY), geometry PS (Geom. PS), attribute XPS (AttrX. 5), attribute YPS (AttrY. PS), geometry SPS (Geometry Sequence PS), attribute X SPS (AttributeX Sequence PS), attribute Y SPS (AttributeY Sequence PS), AU header (AU Header), and GOF header (GOF Header) in codec 1. Values of 11 and greater are reserved in codec 1.

(366) When `pcc_codec_type` is codec 2 (Codec2: second encoding method), values of 0 to 2 of `pcc_nal_unit_type` are assigned to data A (DataA), metadata A (MetaDataA), and metadata B (MetaDataB) in the codec. Values of 3 and greater are reserved in codec 2.

(367) Next, an order of transmission of data will be described. In the following, restrictions on the order of transmission of NAL units will be described.

(368) Multiplexer **4802** transmits NAL units on a GOF basis or on an AU basis. Multiplexer **4802** arranges the GOF header at the top of a GOF, and arranges the AU header at the top of an AU.

(369) In order to allow the decoding device to decode the next AU and the following AUs even when data is lost because of a packet loss or the like, multiplexer **4802** may arrange a sequence parameter set (SPS) in each AU.

(370) When there is a dependence relationship for decoding between encoded data, the decoding device decodes the data of the entity to be referred to and then decodes the data of the referring entity. In order to allow the decoding device to perform decoding in the order of reception without rearranging the data, multiplexer **4802** first transmits the data of the entity to be referred to.

(371) FIG. **40** is a diagram showing examples of the order of transmission of NAL units. FIG. **40** shows three examples, that is, geometry information-first order, parameter-first order, and data-integrated order.

(372) The geometry information-first order of transmission is an example in which information relating to geometry information is transmitted together, and information relating to attribute information is transmitted together. In the case of this order of transmission, the transmission of the information relating to the geometry information ends earlier than the transmission of the information relating to the attribute information.

(373) For example, according to this order of transmission is used, when the decoding device does not decode attribute information, the decoding device may be able to have an idle time since the decoding device can omit decoding of attribute information. When the decoding device is required to decode geometry information early, the decoding device may be able to decode geometry information earlier since the decoding device obtains encoded data of the geometry information earlier.

(374) Note that, although in FIG. 40 the attribute XSPS and the attribute YSPS are integrated and shown as the attribute SPS, the attribute XSPS and the attribute YSPS may be separately arranged.

(375) In the parameter set-first order of transmission, a parameter set is first transmitted, and data is then transmitted.

(376) As described above, as far as the restrictions on the order of transmission of NAL units are met, multiplexer **4802** can transmit NAL units in any order. For example, order identification information may be defined, and multiplexer **4802** may have a function of transmitting NAL units in a plurality of orders. For example, the order identification information for NAL units is stored in the stream PS.

(377) The three-dimensional data decoding device may perform decoding based on the order identification information. The three-dimensional data decoding device may indicate a desired order of transmission to the three-dimensional data encoding device, and the three-dimensional data encoding device (multiplexer **4802**) may control the order of transmission according to the indicated order of transmission.

(378) Note that multiplexer **4802** can generate encoded data having a plurality of functions merged to each other as in the case of the data-integrated order of transmission, as far as the restrictions on the order of transmission are met. For example, as shown in FIG. 40, the GOF header and the AU header may be integrated, or AXPS and AYPS may be integrated. In such a case, an identifier that indicates data having a plurality of functions is defined in `pcc_nal_unit_type`.

(379) In the following, variations of this embodiment will be described. There are levels of PSs, such as a frame-level PS, a sequence-level PS, and a PCC sequence-level PS. Provided that the PCC sequence level is a higher level, and the frame level is a lower level, parameters can be stored in the manner described below.

(380) The value of a default PS is indicated in a PS at a higher level. If the value of a PS at a lower level differs from the value of the PS at a higher level, the value of the PS is indicated in the PS at the lower level. Alternatively, the value of the PS is not described in the PS at the higher level but is described in the PS at the lower level. Alternatively, information indicating whether the value of the PS is indicated in the PS at the lower level, at the higher level, or at both the levels is indicated in both or one of the PS at the lower level and the PS at the higher level. Alternatively, the PS at the lower level may be merged with the PS at the higher level. If the PS at the lower level and the PS at the higher level overlap with each other, multiplexer **4802** may omit transmission of one of the PSs.

(381) Note that encoder **4801** or multiplexer **4802** may divide data into slices or tiles and transmit each of the divided slices or tiles as divided data. The divided data includes information for identifying the divided data, and a parameter used for decoding of the divided data is included in the parameter set. In this case, an identifier that indicates that the data is data relating to a tile or slice or data storing a parameter is defined in `pcc_nal_unit_type`.

(382) In the following, a process relating to order identification information will be described. FIG. 41 is a flowchart showing a process performed by the three-dimensional data encoding device (encoder **4801** and multiplexer **4802**) that involves the order of transmission of NAL units.

(383) First, the three-dimensional data encoding device determines the order of transmission of NAL units (geometry information-first or parameter set-first) (**S4801**). For example, the three-dimensional data encoding device determines the order of transmission based on a specification from a user or an external device (the three-dimensional data decoding device, for example).

(384) If the determined order of transmission is geometry information-first (if “geometry information-first” in **S4802**), the three-dimensional data encoding device sets the order identification information included in the stream PS to geometry information-first (**S4803**). That is, in this case, the order identification information indicates that the NAL units are transmitted in the geometry information-first order. The three-dimensional data encoding device then transmits the NAL units in the geometry information-first order (**S4804**).

(385) On the other hand, if the determined order of transmission is parameter set-first (if

“parameter set-first” in **S4802**), the three-dimensional data encoding device sets the order identification information included in the stream PS to parameter set-first (**S4805**). That is, in this case, the order identification information indicates that the NAL units are transmitted in the parameter set-first order. The three-dimensional data encoding device then transmits the NAL units in the parameter set-first order (**S4806**).

(386) FIG. **42** is a flowchart showing a process performed by the three-dimensional data decoding device that involves the order of transmission of NAL units. First, the three-dimensional data decoding device analyzes the order identification information included in the stream PS (**S4811**).

(387) If the order of transmission indicated by the order identification information is geometry information-first (if “geometry information-first” in **S4812**), the three-dimensional data decoding device decodes the NAL units based on the determination that the order of transmission of the NAL units is geometry information-first (**S4813**).

(388) On the other hand, if the order of transmission indicated by the order identification information is parameter set-first (if “parameter set-first” in **S4812**), the three-dimensional data decoding device decodes the NAL units based on the determination that the order of transmission of the NAL units is parameter set-first (**S4814**).

(389) For example, if the three-dimensional data decoding device does not decode attribute information, in step **S4813**, the three-dimensional data decoding device does not obtain the entire NAL units but can obtain a part of a NAL unit relating to the geometry information and decode the obtained NAL unit to obtain the geometry information.

(390) Next, a process relating to generation of an AU and a GOF will be described. FIG. **43** is a flowchart showing a process performed by the three-dimensional data encoding device (multiplexer **4802**) that relates to generation of an AU and a GOF in multiplexing of NAL units.

(391) First, the three-dimensional data encoding device determines the type of the encoded data (**S4821**). Specifically, the three-dimensional data encoding device determines whether the encoded data to be processed is AU-first data, GOF-first data, or other data.

(392) If the encoded data is GOF-first data (if “GOF-first” in **S4822**), the three-dimensional data encoding device generates NAL units by arranging a GOF header and an AU header at the top of the encoded data belonging to the GOF (**S4823**).

(393) If the encoded data is AU-first data (if “AU-first” in **S4822**), the three-dimensional data encoding device generates NAL units by arranging an AU header at the top of the encoded data belonging to the AU (**S4824**).

(394) If the encoded data is neither GOF-first data nor AU-first data (if “other than GOF-first and AU-first” in **S4822**), the three-dimensional data encoding device generates NAL units by arranging the encoded data to follow the AU header of the AU to which the encoded data belongs (**S4825**).

(395) Next, a process relating to access to an AU and a GOF will be described. FIG. **44** is a flowchart showing a process performed by the three-dimensional data decoding device that involves accessing to an AU and a GOF in demultiplexing of a NAL unit.

(396) First, the three-dimensional data decoding device determines the type of the encoded data included in the NAL unit by analyzing `nal_unit_type` in the NAL unit (**S4831**). Specifically, the three-dimensional data decoding device determines whether the encoded data included in the NAL unit is AU-first data, GOF-first data, or other data.

(397) If the encoded data included in the NAL unit is GOF-first data (if “GOF-first” in **S4832**), the three-dimensional data decoding device determines that the NAL unit is a start position of random access, accesses the NAL unit, and starts the decoding process (**S4833**).

(398) If the encoded data included in the NAL unit is AU-first data (if “AU-first” in **S4832**), the three-dimensional data decoding device determines that the NAL unit is AU-first, accesses the data included in the NAL unit, and decodes the AU (**S4834**).

(399) If the encoded data included in the NAL unit is neither GOF-first data nor AU-first data (if “other than GOF-first and AU-first” in **S4832**), the three-dimensional data decoding device does

not process the NAL unit.

(400) As described above, the three-dimensional data encoding device performs the process shown in FIG. 45. The three-dimensional data encoding device encodes time-series three-dimensional data (point cloud data on a dynamic object, for example). The three-dimensional data includes geometry information and attribute information associated with each time point.

(401) First, the three-dimensional data encoding device encodes the geometry information (S4841). The three-dimensional data encoding device then encodes the attribute information to be processed by referring to the geometry information associated with the same time point as the attribute information to be processed (S4842). Here, as shown in FIG. 37, the geometry information and the attribute information associated with the same time point form an access unit (AU). That is, the three-dimensional data encoding device encodes the attribute information to be processed by referring to the geometry information included in the same access unit as the attribute information to be processed.

(402) In this way, the three-dimensional data encoding device can take advantage of the access unit to facilitate control of reference in encoding. Therefore, the three-dimensional data encoding device can reduce the processing amount of the encoding process.

(403) For example, the three-dimensional data encoding device generates a bitstream including the encoded geometry information (encoded geometry data), the encoded attribute information (encoded attribute data), and information indicating the geometry information of the entity to be referred to when encoding the attribute information to be processed.

(404) For example, the bitstream includes a geometry parameter set (geometry PS) that includes control information for the geometry information associated with each time point and an attribute parameter set (attribute PS) that includes control information for the attribute information associated with each time point.

(405) For example, the bitstream includes a geometry sequence parameter set (geometry SPS) that includes control information that is common to a plurality of pieces of geometry information associated with different time points and attribute sequence parameter set (attribute SPS) that includes control information that is common to a plurality of pieces of attribute information associated with different time points.

(406) For example, the bitstream includes a stream parameter set (stream PS) that includes control information that is common to a plurality of pieces of geometry information associated with different time points and a plurality of pieces of attribute information associated with different time points.

(407) For example, the bitstream includes an access unit header (AU header) that includes control information that is common in an access unit.

(408) For example, the three-dimensional data encoding device performs encoding in such a manner that groups of frames (GOFs) formed by one or more access units can be independently decoded. That is, the GOF is a random access unit.

(409) For example, the bitstream includes a GOF header that includes control information that is common in a GOF.

(410) For example, the three-dimensional data encoding device includes a processor and a memory, and the processor performs the processes described above using the memory.

(411) As described above, the three-dimensional data decoding device performs the process shown in FIG. 46. The three-dimensional data decoding device decodes time-series three-dimensional data (point cloud data on a dynamic object, for example). The three-dimensional data includes geometry information and attribute information associated with each time point. The geometry information and the attribute information associated with the same time point forms an access unit (AU).

(412) First, the three-dimensional data decoding device decodes the bitstream to obtain the geometry information (S4851). That is, the three-dimensional data decoding device generates the geometry information by decoding the encoded geometry information (encoded geometry data)

included in the bitstream.

(413) The three-dimensional data decoding device then decodes the bitstream to obtain the attribute information to be processed by referring to the geometry information associated with the same time point as the attribute information to be processed (S4852). That is, the three-dimensional data decoding device generates the attribute information by decoding the encoded attribute information (encoded attribute data) included in the bitstream. In this process, the three-dimensional data decoding device refers to the decoded geometry information included in the access unit as the attribute information.

(414) In this way, the three-dimensional data decoding device can take advantage of the access unit to facilitate control of reference in decoding. Therefore, the three-dimensional data decoding device can reduce the processing amount of the decoding process.

(415) For example, the three-dimensional data decoding device obtains, from the bitstream, information indicating the geometry information of the entity to be referred to when decoding the attribute information to be processed, and decodes the attribute information to be processed by referring to the geometry information of the entity to be referred to indicated by the obtained information.

(416) For example, the bitstream includes a geometry parameter set (geometry PS) that includes control information for the geometry information associated with each time point and an attribute parameter set (attribute PS) that includes control information for the attribute information associated with each time point. That is, the three-dimensional data decoding device uses the control information included in the geometry parameter set associated with the time point to be intended for processing to decode the geometry information associated with the time point intended for processing, and uses the control information included in the attribute parameter set associated with the time point intended for processing to decode the attribute information associated with the time point intended for processing.

(417) For example, the bitstream includes a geometry sequence parameter set (geometry SPS) that includes control information that is common to a plurality of pieces of geometry information associated with different time points and an attribute sequence parameter set (attribute SPS) that includes control information that is common to a plurality of pieces of attribute information associated with different time points. That is, the three-dimensional data decoding device uses the control information included in the geometry sequence parameter set to decode a plurality of pieces of geometry information associated with different time points, and uses the control information included in the attribute sequence parameter set to decode a plurality of pieces of attribute information associated with different time points.

(418) For example, the bitstream includes a stream parameter set (stream PS) that includes control information that is common to a plurality of pieces of geometry information associated with different time points and a plurality of pieces of attribute information associated with different time points. That is, the three-dimensional data decoding device uses the control information included in the stream parameter set to decode a plurality of pieces of geometry information associated with different time points and a plurality of pieces of attribute information associated with different time points.

(419) For example, the bitstream includes an access unit header (AU header) that includes control information that is common in an access unit. That is, the three-dimensional data decoding device uses the control information included in the access unit header to decode the geometry information and the attribute information included in the access unit.

(420) For example, the three-dimensional data decoding device independently decodes groups of frames (GOFs) formed by one or more access units. That is, the GOF is a random access unit.

(421) For example, the bitstream includes a GOF header that includes control information that is common in a GOF. That is, the three-dimensional data decoding device decodes the geometry information and the attribute information included in the GOF using the control information

included in the GOF header.

(422) For example, the three-dimensional data decoding device includes a processor and a memory, and the processor performs the processes described above using the memory.

Embodiment 5

(423) Next, a configuration of divider **4911** will be described. FIG. **47** is a block diagram illustrating divider **4911**. Divider **4911** includes slice divider **4931**, geometry information tile divider (geometry tile divider) **4932**, and attribute information tile divider (attribute tile divider) **4933**.

(424) Slice divider **4931** generates a plurality of pieces of slice geometry information by dividing geometry information (position (geometry)) into slices. Slice divider **4931** also generates a plurality of pieces of slice attribute information by dividing attribute information (attribute) into slices. Slice divider **4931** also outputs slice additional information (slice MetaData) including information concerning the slice division and information generated in the slice division.

(425) Geometry information tile divider **4932** generates a plurality of pieces of divisional geometry information (a plurality of pieces of tile geometry information) by dividing a plurality of pieces of slice geometry information into tiles. Geometry information tile divider **4932** also outputs geometry tile additional information (geometry tile MetaData) including information concerning the tile division of geometry information and information generated in the tile division of geometry information.

(426) Attribute information tile divider **4933** generates a plurality of pieces of divisional attribute information (a plurality of pieces of tile attribute information) by dividing a plurality of pieces of slice attribute information into tiles. Attribute information tile divider **4933** also outputs attribute tile additional information (attribute tile MetaData) including information concerning the tile division of attribute information and information generated in the tile division of attribute information.

(427) Note that the number of slices or tiles generated by division is equal to or greater than 1. That is, the slice division or tile division may not be performed.

(428) Although an example in which tile division is performed after slice division has been shown here, slice division may be performed after tile division. Alternatively, other units of division may be defined in addition to slice and tile, and the division may be performed based on three or more units of division.

(429) In the following, a method of dividing point cloud data will be described. FIG. **48** is a diagram illustrating an example of slice division and tile division.

(430) First, a method of slice division will be described. Divider **4911** divides three-dimensional point cloud data into arbitrary point clouds in units of slices. In the slice division, divider **4911** does not separate geometry information and attribute information on a point and collectively divides geometry information and attribute information on a point into slices. That is, divider **4911** performs the slice division in such a manner that the geometry information and the attribute information on any point belong to the same slice. Note that the point cloud data can be divided in any number of slices in any manner as far as this requirement is satisfied. The minimum unit of the division is a point. For example, the geometry information and the attribute information are divided into the same number of slices. For example, after the slice division, geometry information corresponding to a three-dimensional point and attribute information corresponding to the three-dimensional point are included in the same slice.

(431) In the slice division, divider **4911** also generates slice additional information, which is additional information concerning the number of slices and the division method. The slice additional information is common to the geometry information and the attribute information. For example, the slice additional information includes information indicating the reference coordinate position or size or the length of a side of a bounding box after the division. The slice additional information also includes information indicating the number of slices, the type of division and the

like.

(432) Next, a method of tile division will be described. Divider **4911** divides data resulting from the slice division into slice geometry information (G slice) and slice attribute information (A slice), and divides each of the slice geometry information and the slice attribute information into tiles.

(433) Note that although FIG. **48** shows an example in which an octree structure is used for the division, the point cloud data may be divided into any number of slices or tiles in any division method.

(434) Divider **4911** may divide the geometry information and the attribute information in different division methods or in the same division method. Divider **4911** may divide a plurality of slices into tiles in different division methods or in the same division method.

(435) In the tile division, divider **4911** also generates tile additional information concerning the number of tiles and the division method. The tile additional information is different between the geometry information and the attribute information (there are geometry tile additional information and attribute tile additional information). For example, the tile additional information includes information indicating the reference coordinate position or size or the length of a side of a bounding box after the division. The tile additional information also includes information indicating the number of tiles, the type of division and the like.

(436) Next, an example of the method of dividing point cloud data into slices or tiles will be described. As a method of slice or tile division, divider **4911** may use a predetermined method or adaptively change the method to be used depending on the point cloud data.

(437) In the slice division, divider **4911** divides a three-dimensional space without separating the geometry information and the attribute information. For example, divider **4911** determines the shape of an object and divides the three-dimensional space into slices based on the shape of the object. For example, divider **4911** extracts an object, such as a tree or a building, and performs the division on an object basis. For example, divider **4911** performs the slice division in such a manner that the whole of one or more objects is included in one slice. Alternatively, divider **4911** may divide one object into a plurality of slices.

(438) In this case, the encoding device may use a different encoding method for each slice. For example, the encoding device may use a high-quality compression method for a particular object or some particular objects. In that case, the encoding device may store information indicating the encoding method for each slice in the additional information (metadata).

(439) Divider **4911** may also perform the slice division in such a manner that each slice corresponds to a predetermined coordinate space, based on map information or the geometry information.

(440) In the tile division, divider **4911** independently divides the geometry information and the attribute information. For example, divider **4911** divides a slice into tiles depending on the data amount or processing amount. For example, divider **4911** determines whether the data amount of a slice (the number of three-dimensional points included in a slice, for example) is greater than a predetermined threshold. If the data amount of the slice is greater than the threshold, divider **4911** divides the slice into tiles. If the data amount of the slice is smaller than the threshold, divider **4911** does not divide the slice into tiles.

(441) For example, divider **4911** divides a slice into tiles in such a manner that the processing amount or processing time of the decoding device falls within a predetermined range (that is, is equal to or less than a predetermined value). In this way, the processing amount for each tile of the decoding device is made constant, and the distributed processing in the decoding device is facilitated.

(442) When the processing amount for the geometry information and the processing amount for the attribute information are different, for example, the processing amount for the geometry information is greater than the processing amount for the attribute information, divider **4911** divides the geometry information into more tiles than the attribute information.

(443) Depending on the content, if the decoding device can decode and display the geometry information before decoding and displaying the attribute information, for example, divider **4911** may divide the geometry information into more tiles than the attribute information. In that case, the decoding device can increase the parallelism of the processing of the geometry information compared with the processing of the attribute information, and therefore can more quickly process the geometry information than the attribute information.

(444) Note that the decoding device does not have to process the slices or tiles of data in parallel, and may determine whether to process the slices or tiles of data in parallel or not based on the number or capacity of the decoding processors.

(445) By performing the divisions as described above, adaptive encoding can be realized depending on the content or object. In addition, a parallel decoding process can be realized. In this way, the flexibility of the point cloud data encoding system or point cloud data decoding system is improved.

(446) FIG. **49** is a diagram illustrating an example of a slice division pattern and a tile division pattern. In the drawing, DU represents a data unit (DataUnit), and shows a tile or slice of data. Each DU includes a slice index (SliceIndex) and a tile index (TileIndex). In the drawing, a superscript of DU indicates a slice index, and a subscript of DU indicates a tile index.

(447) In pattern 1, in the slice division, the geometry information and the attribute information are divided into the same number of G slices and A slices in the same division method. In the tile division, the G slice and the A slice are divided into different numbers of tiles in different division methods. The plurality of G slices is divided into the same number of tiles in the same division method. The plurality of A slices is divided into the same number of tiles in the same division method.

(448) In pattern 2, in the slice division, the geometry information and the attribute information are divided into the same number of G slices and A slices in the same division method. In the tile division, the G slice and the A slice are divided into different numbers of tiles in different division methods. The plurality of G slices are divided into different numbers of tiles in different division methods. The plurality of A slices are divided into different numbers of tiles in different division methods.

Embodiment 6

(449) In order to divide point cloud data into tiles and slices and efficiently encode or decode the divisional data, an appropriate control is needed on the encoder side and the decoder side. By making the encoding and decoding of each piece of divisional data independent, rather than dependent, from the other pieces of divisional data, a multi-thread or multi-core processor can be used to process the pieces of divisional data in the respective threads/cores in parallel, and the performance is improved.

(450) There are various methods of dividing point cloud data into tiles and slices. For example, there is a method of dividing point cloud data based on an attribute of an object, such as a road surface, of point cloud data or a characteristic, such as color information such as green, of point cloud data.

(451) CABAC is an abbreviation of context-based adaptive binary arithmetic coding, which is an encoding method that realizes an arithmetic encoding (entropy encoding) with high compression ratio by increasing the probability precision by successively updating a context (a model for estimating the probability of occurrence of an input binary symbol) based on the encoded information.

(452) In order to process pieces of divisional data such as tiles or slices in parallel, each piece of divisional data needs to be independently encoded or decoded. In order to make CABAC for the pieces of divisional data independent from each other, CABAC needs to be initialized at the top of each piece of divisional data. However, there is no mechanism therefor.

(453) A CABAC initialization flag is used to initialize CABAC in CABAC encoding and decoding.

(454) FIG. 50 is a flowchart of a process of initializing CABAC in response to a CABAC initialization flag.

(455) The three-dimensional data encoding device or three-dimensional data decoding device determines whether the CABAC initialization flag is 1 or not in encoding or decoding (S5201).

(456) When the CABAC initialization flag is 1 (if Yes in S5201), the three-dimensional data encoding device or three-dimensional data decoding device initializes a CABAC encoder/decoder to a default state (S5202), and continues the encoding or decoding.

(457) When the CABAC initialization flag is not 1 (if No in S5201), the three-dimensional data encoding device or three-dimensional data decoding device does not perform the initialization, and continues the encoding or decoding.

(458) That is, when initializing CABAC, cabac_init_flag is set to 1, and the CABAC encoder or CABAC decoder is initialized or re-initialized. When initializing CABAC, an initial value (default state) of a context used for the CABAC process is set.

(459) An encoding process will be described. FIG. 51 is a block diagram illustrating a configuration of first encoder 5200 included in the three-dimensional data encoding device according to this embodiment. FIG. 52 is a block diagram illustrating a configuration of divider 5201 according to this embodiment. FIG. 53 is a block diagram illustrating a configuration of geometry information encoder 5202 and attribute information encoder 5203 according to this embodiment.

(460) First encoder 5200 generates encoded data (encoded stream) by encoding point cloud data in a first encoding method (geometry-based PCC (GPCC)). First encoder 5200 includes divider 5201, a plurality of geometry information encoders 5202, a plurality of attribute information encoders 5203, additional information encoder 5204, and multiplexer 5205.

(461) Divider 5201 generates a plurality of pieces of divisional data by dividing point cloud data. Specifically, divider 5201 generates a plurality of pieces of divisional data by dividing a space of point cloud data into a plurality of subspaces. Here, a subspace is a combination of tiles or slices or a combination of tiles and slices. More specifically, point cloud data includes geometry information, attribute information, and additional information. Divider 5201 divides geometry information into a plurality of pieces of divisional geometry information, and divides attribute information into a plurality of pieces of divisional attribute information. Divider 5201 also generates additional information concerning the division.

(462) As illustrated in FIG. 52, divider 5201 includes tile divider 5211 and slice divider 5212. For example, tile divider 5211 divides a point cloud into tiles. Tile divider 5211 may determine a quantization value used for each divisional tile as tile additional information.

(463) Slice divider 5212 further divides a tile obtained by tile divider 5211 into slices. Slice divider 5212 may determine a quantization value used for each divisional slice as slice additional information.

(464) The plurality of geometry information encoders 5202 generate a plurality of pieces of encoded geometry information by encoding a plurality of pieces of divisional geometry information. For example, the plurality of geometry information encoders 5202 processes a plurality of pieces of divisional geometry information in parallel.

(465) As illustrated in FIG. 53, geometry information encoder 5202 includes CABAC initializer 5221 and entropy encoder 5222. CABAC initializer 5221 initializes or re-initializes CABAC in response to a CABAC initialization flag. Entropy encoder 5222 encodes divisional geometry information according to CABAC.

(466) The plurality of attribute information encoders 5203 generate a plurality of pieces of encoded attribute information by encoding a plurality of pieces of divisional attribute information. For example, the plurality of attribute information encoders 5203 process a plurality of pieces of divisional attribute information in parallel.

(467) As illustrated in FIG. 53, attribute information encoder 5203 includes CABAC initializer

5231 and entropy encoder **5232**. CABAC initializer **5231** initializes or re-initializes CABAC in response to a CABAC initialization flag. Entropy encoder **5232** encodes divisional attribute information according to CABAC.

(468) Additional information encoder **5204** generates encoded additional information by encoding additional information included in the point cloud data and additional information concerning the data division generated in the division by divider **5201**.

(469) Multiplexer **5205** generates encoded data (encoded stream) by multiplexing a plurality of pieces of encoded geometry information, a plurality of pieces of encoded attribute information, and encoded additional information, and transmits the generated encoded data. The encoded additional information is used for decoding.

(470) Note that, although FIG. **51** shows an example in which there are two geometry information encoders **5202** and two attribute information encoders **5203**, the number of geometry information encoders **5202** and the number of attribute information encoders **5203** may be one, or three or more. The plurality of pieces of divisional data may be processed in parallel in the same chip, such as by a plurality of cores of a CPU, processed in parallel by cores of a plurality of chips, or processed in parallel by a plurality of cores of a plurality of chips.

(471) Next, a decoding process will be described. FIG. **54** is a block diagram illustrating a configuration of first decoder **5240**. FIG. **55** is a block diagram illustrating a configuration of geometry information decoder **5242** and attribute information decoder **5243**.

(472) First decoder **5240** reproduces point cloud data by decoding encoded data (encoded stream) generated by encoding the point cloud data in the first encoding method (GPCC). First decoder **5240** includes demultiplexer **5241**, a plurality of geometry information decoders **5242**, a plurality of attribute information decoders **5243**, additional information decoder **5244**, and combiner **5245**.

(473) Demultiplexer **5241** generates a plurality of pieces of encoded geometry information, a plurality of pieces of encoded attribute information, and encoded additional information by demultiplexing encoded data (encoded stream).

(474) The plurality of geometry information decoders **5242** generates a plurality of pieces of quantized geometry information by decoding a plurality of pieces of encoded geometry information. For example, the plurality of geometry information decoders **5242** process a plurality of pieces of encoded geometry information in parallel.

(475) As illustrated in FIG. **55**, geometry information decoder **5242** includes CABAC initializer **5251** and entropy decoder **5252**. CABAC initializer **5251** initializes or re-initializes CABAC in response to a CABAC initialization flag. Entropy decoder **5252** decodes geometry information according to CABAC.

(476) The plurality of attribute information decoders **5243** generate a plurality of pieces of divisional attribute information by decoding a plurality of pieces of encoded attribute information. For example, the plurality of attribute information decoders **5243** process a plurality of pieces of encoded attribute information in parallel.

(477) As illustrated in FIG. **55**, attribute information decoder **5243** includes CABAC initializer **5261** and entropy decoder **5262**. CABAC initializer **5261** initializes or re-initializes CABAC in response to a CABAC initialization flag. Entropy decoder **5262** decodes attribute information according to CABAC.

(478) The plurality of additional information decoders **5244** generate additional information by decoding encoded additional information.

(479) Combiner **5245** generates geometry information by combining a plurality of pieces of divisional geometry information using additional information. Combiner **5245** generates attribute information by combining a plurality of pieces of divisional attribute information using additional information. For example, combiner **5245** first generates point cloud data associated with a tile by combining decoded point cloud data associated with slices using slice additional information. Combiner **5245** then reproduces the original point cloud data by combining point cloud data

associated with tiles using tile additional information.

(480) Note that, although FIG. 54 shows an example in which there are two geometry information decoders 5242 and two attribute information decoders 5243, the number of geometry information decoders 5242 and the number of attribute information decoders 5243 may be one, or three or more. The plurality of pieces of divisional data may be processed in parallel in the same chip, such as by a plurality of cores of a CPU, processed in parallel by cores of a plurality of chips, or processed in parallel by a plurality of cores of a plurality of chips.

(481) FIG. 56 is a flowchart illustrating an example of a process associated with the initialization of CABAC in the encoding of geometry information or the encoding of attribute information.

(482) First, the three-dimensional data encoding device determines, for each slice, whether or not to initialize CABAC in the encoding of geometry information for the slice based on a predetermined condition (S5201).

(483) When it is determined to initialize CABAC (if Yes in S5202), the three-dimensional data encoding device determines a context initial value used for the encoding of geometry information (S5203). The context initial value is set by considering encoding characteristics. The initial value may be a predetermined value or may be adaptively determined depending on the characteristics of data in the slice.

(484) The three-dimensional data encoding device then sets the CABAC initialization flag for geometry information to be 1, and sets the context initial value (S5204). When initializing CABAC, the initialization process is performed using the context initial value in the encoding of geometry information.

(485) On the other hand, when it is determined not to initialize CABAC (if No in S5202), the three-dimensional data encoding device sets the CABAC initialization flag for geometry information to be 0 (S5205).

(486) The three-dimensional data encoding device then determines, for each slice, whether or not to initialize CABAC in the encoding of attribute information for the slice based on a predetermined condition (S5206).

(487) When it is determined to initialize CABAC (if Yes in S5207), the three-dimensional data encoding device determines a context initial value used for the encoding of attribute information (S5208). The context initial value is set by considering encoding characteristics. The initial value may be a predetermined value or may be adaptively determined depending on the characteristics of data in the slice.

(488) The three-dimensional data encoding device then sets the CABAC initialization flag for attribute information to be 1, and sets the context initial value (S5209). When initializing CABAC, the initialization process is performed using the context initial value in the encoding of attribute information.

(489) On the other hand, when it is determined not to initialize CABAC (if No in S5207), the three-dimensional data encoding device sets the CABAC initialization flag for attribute information to be 0 (S5210).

(490) Note that, in the flowchart of FIG. 56, the processing concerning geometry information and the processing concerning attribute information may be performed in reverse order or in parallel.

(491) Note that, although the flowchart of FIG. 56 shows a slice-based process as an example, a tile-based process or a process on a basis of other data units can be performed in the same manner as the slice-based process. That is, slice in the flowchart of FIG. 56 can be replaced with tile or other data units.

(492) The predetermined condition for the geometry information and the predetermined condition for the attribute information may be the same condition or different conditions.

(493) FIG. 57 is a diagram illustrating an example of timings of CABAC initialization for point cloud data in the form of a bitstream.

(494) Point cloud data includes geometry information and zero or more pieces of attribute

information. That is, point cloud data may include no attribute information or include a plurality of pieces of attribute information.

(495) For example, as attribute information on a three-dimensional point, point cloud data may include color information, may include color information and reflection information, or may include one or more pieces of color information each linked to one or more pieces of point-of-view information.

(496) In any configuration, the method described in this embodiment can be applied.

(497) Next, a condition for determination of whether to initialize CABAC will be described.

(498) It may be determined to initialize CABAC in the encoding of geometry information or attribute information when any of the conditions described below is satisfied.

(499) For example, CABAC may be initialized at the leading data of geometry information or attribute information (each piece of attribute information if there is a plurality of pieces of attribute information). For example, CABAC may be initialized at the top of data forming a PCC frame that can be singly decoded. That is, as illustrated in part (a) of FIG. 57, if PCC frames may be decoded on a frame basis, CABAC can be initialized at the leading data of a PCC frame.

(500) For example, as illustrated in part (b) of FIG. 57, if frames cannot be singly decoded, such as when an inter-prediction is used between PCC frames, CABAC may be initialized at the leading data of a random access unit (GOF, for example).

(501) For example, as illustrated in part (c) of FIG. 57, CABAC may be initialized at the top of one or more pieces of divisional slice data, at the top of one or more pieces of divisional tile data, or at the top of other divisional data.

(502) Although part (c) of FIG. 57 shows tiles as an example, this description holds true for slices. CABAC may be always initialized at the top of a tile or slice or may not be always initialized at the top of a tile or slice.

(503) FIG. 58 is a diagram illustrating a configuration of encoded data and a method of storing the encoded data into a NAL unit.

(504) Initialization information may be stored in a header of encoded data or in metadata. The initialization information may also be stored in both the header and the metadata. The initialization information is `cabac_init_flag`, a CABAC initial value, or an index of a table capable of identifying an initial value.

(505) In this embodiment, “metadata” in a description that something is stored in metadata can be replaced with “header of encoded data” or vice versa.

(506) When the initialization information is stored in the header of encoded data, the initialization information may be stored in the first NAL unit in the encoded data, for example. Initialization information on the encoding of geometry information is stored in geometry information, and initialization information on the encoding of attribute information is stored in attribute information.

(507) `cabac_init_flag` for the encoding of attribute information and `cabac_init_flag` for the encoding of geometry information may be set to be the same value or different values. When the flags are set to be the same value, `cabac_init_flag` may be shared for geometry information and attribute information. When the flags are set to be different values, `cabac_init_flag` for geometry information and `cabac_init_flag` for attribute information indicate different values.

(508) The initialization information for geometry information and the initialization information for attribute information may be stored in common metadata, at least one of individual metadata of geometry information and individual metadata of attribute information, or both the common metadata and the individual metadata. A flag may be used which indicates in which of the individual metadata for geometry information, the individual metadata for attribute information, and the common metadata the initialization information is stored.

(509) FIG. 59 is a flowchart illustrating an example of a process associated with the initialization of CABAC in the decoding of geometry information or the decoding of attribute information.

(510) The three-dimensional data decoding device analyzes encoded data to obtain a CABAC

initialization flag for geometry information, a CABAC initialization flag for attribute information, and a context initial value (S5211).

(511) The three-dimensional data decoding device then determines whether the CABAC initialization flag for geometry information is 1 or not (S5512).

(512) When the CABAC initialization flag for geometry information is 1 (if Yes in S5212), the three-dimensional data decoding device initializes the CABAC decoding for the encoded geometry information using the context initial value in the encoding of the geometry information (S5213).

(513) On the other hand, when the CABAC initialization flag for geometry information is 0 (if No in S5212), the three-dimensional data decoding device does not initialize the CABAC decoding for the encoded geometry information (S5214).

(514) The three-dimensional data decoding device then determines whether the CABAC initialization flag for attribute information is 1 or not (S5215). When the CABAC initialization flag for attribute information is 1 (if Yes in S5215), the three-dimensional data decoding device initializes the CABAC decoding for the encoded attribute information using the context initial value in the encoding of the attribute information (S5216).

(515) On the other hand, when the CABAC initialization flag for attribute information is 0 (if No in S5215), the three-dimensional data decoding device does not initialize the CABAC decoding for the encoded attribute information (S5217).

(516) Note that, in the flowchart of FIG. 59, the processing concerning geometry information and the processing concerning attribute information may be performed in reverse order or in parallel.

(517) Note that the flowchart of FIG. 59 can be applied to any of the case of slice division and the case of tile division.

(518) Next, a flow of a process of encoding point cloud data and a flow of a process of decoding point cloud data according to this embodiment will be described. FIG. 60 is a flowchart of a process of encoding point cloud data according to this embodiment.

(519) First, the three-dimensional data encoding device determines a division method to be used (S5221). The division method includes a determination of whether to perform tile division or not and a determination of whether to perform slice division or not. The division method may include the number of tiles or slices in the case where tile division or slice division is performed, and the type of division, for example. The type of division is a scheme based on an object shape, a scheme based on map information or geometry information, or a scheme based on a data amount or processing amount, for example. The division method may be determined in advance.

(520) When tile division is to be performed (if Yes in S5222), the three-dimensional data encoding device generates a plurality of pieces of tile geometry information and a plurality of pieces of tile attribute information by dividing the geometry information and the attribute information on a tile basis (S5223). The three-dimensional data encoding device also generates tile additional information concerning the tile division.

(521) When slice division is to be performed (if Yes in S5224), the three-dimensional data encoding device generates a plurality of pieces of divisional geometry information and a plurality of pieces of divisional attribute information by dividing the plurality of pieces of tile geometry information and the plurality of pieces of tile attribute information (or the geometry information and the attribute information) (S5225). The three-dimensional data encoding device also generates geometry slice additional information and attribute slice additional information concerning the slice division.

(522) The three-dimensional data encoding device then generates a plurality of pieces of encoded geometry information and a plurality of pieces of encoded attribute information by encoding each of the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information (S5226). The three-dimensional data encoding device also generates dependency information.

(523) The three-dimensional data encoding device then generates encoded data (encoded stream)

by integrating (multiplexing) the plurality of pieces of encoded geometry information, the plurality of pieces of encoded attribute information and the additional information into a NAL unit (S5227). The three-dimensional data encoding device also transmits the generated encoded data.

(524) FIG. **61** is a flowchart illustrating an example of a process of determining the value of the CABAC initialization flag and updating additional information in the tile division (S5223) and the slice division (S5525).

(525) In steps S5223 and S5225, tile geometry information and tile attribute information and/or slice geometry information and slice attribute information may be independently divided in respective manners, or may be collectively divided in a common manner. In this way, additional information divided on a tile basis and/or on a slice basis is generated.

(526) In these steps, the three-dimensional data encoding device determines whether to set the CABAC initialization flag to 1 or 0 (S5231).

(527) The three-dimensional data encoding device then updates the additional information to include the determined CABAC initialization flag (S5232).

(528) FIG. **62** is a flowchart illustrating an example of a process of initializing CABAC in the processing of encoding (S5226).

(529) The three-dimensional data encoding device determines whether the CABAC initialization flag is 1 or not (S5241).

(530) When the CABAC initialization flag is 1 (if Yes in S5241), the three-dimensional data encoding device re-initializes the CABAC encoder to the default state (S5242).

(531) The three-dimensional data encoding device then continues the encoding process until a condition for stopping the encoding process is satisfied, such as until there is no data to be encoded (S5243).

(532) FIG. **63** is a flowchart illustrating a process of decoding point cloud data according to this embodiment. First, the three-dimensional data decoding device determines the division method by analyzing additional information (tile additional information, geometry slice additional information, and attribute slice additional information) concerning the division method included in encoded data (encoded stream) (S5251). The division method includes a determination of whether to perform tile division or not and a determination of whether to perform slice division or not. The division method may include, for example, the number of tiles or slices and the type of division in the case where tile division or slice division is performed.

(533) The three-dimensional data decoding device then generates divisional geometry information and divisional attribute information by decoding a plurality of pieces of encoded geometry information and a plurality of pieces of encoded attribute information included in the encoded data using dependency information included in the encoded data (S5252).

(534) If the additional information indicates that slice division has been performed (if Yes in S5253), the three-dimensional data decoding device generates a plurality of pieces of tile geometry information and a plurality of pieces of tile attribute information by combining the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information based on the geometry slice additional information and the attribute slice additional information (S5254).

(535) If the additional information indicates that tile division has been performed (if Yes in S5255), the three-dimensional data decoding device generates geometry information and attribute information by combining the plurality of pieces of tile geometry information and the plurality of pieces of tile attribute information (the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information) based on the tile additional information (S5256).

(536) FIG. **64** is a flowchart illustrating an example of a process of initializing the CABAC decoder in the combining (S5254) of information divided into slices and the combining (S5256) of information divided into tiles.

(537) Pieces of slice geometry information and pieces of slice attribute information or pieces of tile geometry information or pieces of tile attribute information may be combined in respective manners or in the same manner.

(538) The three-dimensional data decoding device obtains the CABAC initialization flag by decoding the additional information in the encoded stream.

(539) The three-dimensional data decoding device then determines whether the CABAC initialization flag is 1 or not (S5262).

(540) When the CABAC initialization flag is 1 (if Yes in S5262), the three-dimensional data decoding device re-initializes the CABAC decoder to the default state (S5263).

(541) On the other hand, when the CABAC initialization flag is not 1 (if No in S5262), the three-dimensional data decoding device does not re-initialize the CABAC decoder and proceeds to step S5264.

(542) The three-dimensional data decoding device then continues the decoding process until a condition for stopping the decoding process is satisfied, such as until there is no data to be decoded (S5264).

(543) Next, other conditions concerning the determination of whether to initialize CABAC will be described.

(544) Whether to initialize the encoding of geometry information or the encoding of attribute information may be determined by considering the coding efficiency on a basis of data units, such as tiles or slices. In that case, CABAC may be initialized at the leading data of a tile or slice that satisfies a predetermined condition.

(545) Next, conditions concerning the determination of whether to initialize CABAC in the encoding of geometry information will be described.

(546) For example, the three-dimensional data encoding device may determine the density of point cloud data for each slice, that is, the number of points per unit area belonging to each slice, compare the data density of the slice with the data density of another slice, and determine that the coding efficiency is better when CABAC is not initialized and determine not to initialize CABAC if the variation of the data density satisfies a predetermined condition. On the other hand, if the variation of the data density does not satisfy the predetermined condition, the three-dimensional data encoding device may determine that the coding efficiency is better when CABAC is initialized, and determine to initialize CABAC.

(547) Here, “another slice” may be the preceding slice in the decoding order or a spatially neighboring slice, for example. The three-dimensional data encoding device may not perform the comparison of the data density with that of another slice and may determine whether to initialize CABAC based on whether the data density of the slice is a predetermined data density or not.

(548) When it is determined to initialize CABAC, the three-dimensional data encoding device determines the context initial value used for the encoding of geometry information. The context initial value is set at a value that provides good encoding characteristics in response to the data density. The three-dimensional data encoding device may retain an initial value table for the data density in advance and selects an optimal initial value from the table.

(549) Note that the three-dimensional data encoding device may determine whether to initialize CABAC based on the number of points, the distribution of points, or the imbalance of points, for example, rather than based on the density of a slice described above as an example. Alternatively, the three-dimensional data encoding device may determine whether to initialize CABAC based on a feature quantity or the number of feature points obtained from information on points or based on a recognized object. In that case, a determination criterion may be retained in a memory in the form of a table that associates the determination criterion with a feature quantity or the number of feature points obtained from information on points or an object recognized based on information on points.

(550) For example, the three-dimensional data encoding device may determine an object associated with geometry information of map information and determine whether to initialize CABAC based

on the object based on the geometry information. Alternatively, the three-dimensional data encoding device may determine whether to initialize CABAC based on information or a feature quantity obtained by projecting three-dimensional data onto a two-dimensional plane.

(551) Next, conditions concerning the determination of whether to initialize CABAC in the encoding of attribute information will be described.

(552) For example, the three-dimensional data encoding device may compare a color characteristic of the relevant slice with the color characteristic of the preceding slice, and determine that the coding efficiency is better when CABAC is not initialized and determine not to initialize CABAC if the variation of the color characteristic satisfies a predetermined condition. On the other hand, if the variation of the color characteristic does not satisfy the predetermined condition, the three-dimensional data encoding device may determine that the coding efficiency is better when CABAC is initialized, and determine to initialize CABAC. The color characteristic is luminance, chromaticity, or chroma, a histogram thereof, or color continuity, for example.

(553) Here, “another slice” may be the preceding slice in the decoding order or a spatially neighboring slice, for example. The three-dimensional data encoding device may not perform the comparison of the data density with that of another slice and may determine whether to initialize CABAC based on whether the data density of the slice is a predetermined data density or not.

(554) When it is determined to initialize CABAC, the three-dimensional data encoding device determines the context initial value used for the encoding of attribute information. The context initial value is set at a value that provides good encoding characteristics in response to the data density. The three-dimensional data encoding device may retain an initial value table for the data density in advance and select an optimal initial value from the table.

(555) When the attribute information is reflectance, the three-dimensional data encoding device may determine whether to initialize CABAC based on reflectance-based information.

(556) When a three-dimensional point has a plurality of pieces of attribute information, the three-dimensional data encoding device may independently determine initialization information for each piece of attribute information based on the piece of attribute information, may determine initialization information for the plurality of pieces of attribute information based on one of the pieces of attribute information, or may determine initialization information for the plurality of pieces of attribute information using a plurality of pieces of attribute information.

(557) Although an example has been described in which the initialization information for geometry information is determined based on the geometry information, and the initialization information for attribute information is determined based on the attribute information, the initialization information for geometry information and attribute information may be determined based on the geometry information, based on the attribute information, or based on both the geometry information and the attribute information.

(558) The three-dimensional data encoding device may determine initialization information based on a result of simulation of the coding efficiency performed by turning on and off `cabac_init_flag` or selecting one or more initial values from an initial value table, for example.

(559) When the data division method into tiles, slices or the like is determined based on geometry information or attribute information, the three-dimensional data encoding device may determine initialization information based on the same information as information based on the determination of the division method.

(560) FIG. 65 is a diagram illustrating an example of tiles and slices.

(561) For example, slices in one tile having part of PCC data are recognized as indicated by legends. The CABAC initialization flag can be used to determine whether re-initialization of a context is needed or not in successive slices. For example, in FIG. 65, when one tile includes slice data divided on a basis of objects (such as a moving body, a sidewalk, a building, a tree or other objects), the CABAC initialization flags for slices of a moving body, a sidewalk, and a tree are set to be 1, and the CABAC initialization flags for slices of a building and other objects are set to be 0.

This means that, if the sidewalk and the building may be both dense permanent structures and have similar coding efficiencies, the coding efficiency may be able to be improved by avoiding re-initialization of CABAC between the slices for the sidewalk and the building. On the other hand, if the building and the tree may be significantly different in density and coding efficiency, the coding efficiency may be able to be improved by initializing CABAC between the slices for the building and the tree.

(562) FIG. **66** is a flowchart illustrating an example of the method of determining whether to initialize CABAC and determining a context initial value.

(563) First, the three-dimensional data encoding device divides point cloud data into slices based on an object determined from geometry information (S5271).

(564) The three-dimensional data encoding device then determines, for each slice, whether to initialize CABAC for the encoding of geometry information and the encoding of attribute information based on the data density of the object of the slice (S5272). In other words, the three-dimensional data encoding device determines CABAC initialization information (CABAC initialization flag) for the encoding of geometry information and the encoding of attribute information based on the geometry information. The three-dimensional data encoding device determines an initialization with high coding efficiency based on the point cloud data density, for example. The CABAC initialization information may be indicated by `cabac_init_flag` that is common to the geometry information and the attribute information.

(565) When it is determined to initialize CABAC (if Yes in S5273), the three-dimensional data encoding device determines a context initial value for the encoding of geometry information (S5274).

(566) The three-dimensional data encoding device then determines a context initial value for the encoding of attribute information (S5275).

(567) The three-dimensional data encoding device then sets the CABAC initialization flag for geometry information to be 1, sets the context initial value for geometry information, sets the CABAC initialization flag for attribute information to be 1, and sets the context initial value for attribute information (S5276). Note that when initializing CABAC, the three-dimensional data encoding device performs the initialization process using a context initial value in each of the encoding of geometry information and the encoding of attribute information.

(568) On the other hand, when it is determined not to initialize CABAC (if No in S5273), the three-dimensional data encoding device sets the CABAC initialization flag for geometry information to be 0, and sets the CABAC initialization flag for attribute information to be 0 (S5277).

(569) FIG. **67** is a diagram illustrating an example of a case where a map, which is a top view of point cloud data obtained by LiDAR, is divided into tiles. FIG. **68** is a flowchart illustrating another example of the method of determining whether to initialize CABAC and determining a context initial value.

(570) In large-scale map data, the three-dimensional data encoding device divides point cloud data into one or more tiles based on geometry information in a two-dimensional top-view division manner (S5281). The three-dimensional data encoding device may divide point cloud data into square areas as illustrated in FIG. **67**, for example. The three-dimensional data encoding device may also divide point cloud data into tiles of different shapes or sizes. The division into tiles may be performed in one or more methods determined in advance or may be adaptively performed.

(571) The three-dimensional data encoding device then determines an object in each tile, and determines whether to initialize CABAC in the encoding of geometry information for the tile or the encoding of attribute information for the tile (S5282). Note that, in the division into slices, the three-dimensional data encoding device recognizes an object (a tree, a human being, a moving body, or a building), and determines whether to perform the slice division and determine an initial value based on the object.

(572) When it is determined to initialize CABAC (if Yes in S5283), the three-dimensional data

encoding device determines a context initial value for the encoding of geometry information (S5284).

(573) The three-dimensional data encoding device then determines a context initial value for the encoding of attribute information (S5285).

(574) In steps S5284 and S5285, an initial value for a tile having particular encoding characteristics may be stored as the initial value and used as an initial value for a tile having the same encoding characteristics.

(575) The three-dimensional data encoding device then sets the CABAC initialization flag for geometry information to be 1, sets the context initial value for geometry information, sets the CABAC initialization flag for attribute information to be 1, and sets the context initial value for attribute information (S5286). Note that when initializing CABAC, the three-dimensional data encoding device performs the initialization process using a context initial value in each of the encoding of geometry information and the encoding of attribute information.

(576) On the other hand, when it is determined not to initialize CABAC (if No in S5283), the three-dimensional data encoding device sets the CABAC initialization flag for geometry information to be 0, and sets the CABAC initialization flag for attribute information to be 0 (S5287).

Embodiment 7

(577) Next, a quantization parameter will be described.

(578) In order to divide point cloud data based on characteristics and positions concerning the point cloud data, a slice and a tile are used. Here, a different quality may be required for each of the pieces of divisional point cloud data, because of hardware restrictions or requirements for real-time processing, for example. For example, when encoding point cloud data by dividing the point cloud data into slices on an object basis, slice data including a plant is less important, so that the resolution (quality) of the slice data can be decreased by quantization. On the other hand, the resolution (quality) of important slice data can be increased by setting the quantization value at a lower value. A quantization parameter is used to enable such a control of quantization value.

(579) Here, data to be quantized, a scale used for the quantization, and quantized data, which is the result of calculation by the quantization, are expressed by Equations G1 and G2 below.

quantized data=data/scale (Equation G1)

data=quantized data*scale (Equation G2)

(580) FIG. 69 is a diagram for describing a process performed by quantizer 5323 that quantizes data and inverse quantizer 5333 that inverse-quantizes quantized data.

(581) Quantizer 5323 quantizes data using a scale. That is, quantizer 5323 calculates quantized data, which is data quantized, by performing a process according to Equation G1.

(582) Inverse quantizer 5333 inverse-quantizes quantized data using the scale. That is, inverse quantizer calculates inverse-quantized quantized data by performing a process according to Equation G2.

(583) The scale and the quantization value (quantization parameter (QP) value) are expressed by Equation G3 below.

quantization value(QP value)=log(scale) (Equation G3)

quantization value(QP value)=default value(reference value)+quantization delta(difference information) (Equation G4)

(584) These parameters are generically referred to as a quantization parameter.

(585) For example, as illustrated in FIG. 70, a quantization value is a value with respect to a default value, and is calculated by adding a quantization delta to the default value. If the quantization value is smaller than the default value, the quantization delta is a negative value. If the quantization value is greater than the default value, the quantization delta is a positive value. If the quantization value is equal to the default value, the quantization delta is 0. When the quantization delta is 0, the quantization delta can be omitted.

(586) An encoding process will be described. FIG. 71 is a block diagram illustrating a

configuration of first encoder **5300** included in the three-dimensional data encoding device according to this embodiment. FIG. **72** is a block diagram illustrating a configuration of divider **5301** according to this embodiment. FIG. **73** is a block diagram illustrating a configuration of geometry information encoder **5302** and attribute information encoder **5303** according to this embodiment.

(587) First encoder **5300** generates encoded data (encoded stream) by encoding point cloud data in a first encoding method (geometry-based PCC (GPCC)). First encoder **5300** includes divider **5301**, a plurality of geometry information encoders **5302**, a plurality of attribute information encoders **5303**, additional information encoder **5304**, and multiplexer **5305**.

(588) Divider **5301** generates a plurality of pieces of divisional data by dividing point cloud data. Specifically, divider **5301** generates a plurality of pieces of divisional data by dividing a space of point cloud data into a plurality of subspaces. Here, a subspace is a combination of tiles or slices, or a combination of tiles and slices. More specifically, point cloud data includes geometry information, attribute information, and additional information. Divider **5301** divides geometry information into a plurality of pieces of divisional geometry information, and divides attribute information into a plurality of pieces of divisional attribute information. Divider **5301** also generates additional information concerning the division.

(589) As illustrated in FIG. **72**, divider **5301** includes tile divider **5311** and slice divider **5312**. For example, tile divider **5311** divides a point cloud into tiles. Tile divider **5311** may determine a quantization value used for each divisional tile as tile additional information.

(590) Slice divider **5312** further divides a tile obtained by tile divider **5311** into slices. Slice divider **5312** may determine a quantization value used for each divisional slice as slice additional information.

(591) The plurality of geometry information encoders **5302** generate a plurality of pieces of encoded geometry information by encoding a plurality of pieces of divisional geometry information. For example, the plurality of geometry information encoders **5302** process a plurality of pieces of divisional geometry information in parallel.

(592) As illustrated in FIG. **73**, geometry information encoder **5302** includes quantization value calculator **5321** and entropy encoder **5322**. Quantization value calculator **5321** generates a quantization value (quantization parameter) of divisional geometry information to be encoded. Entropy encoder **5322** calculates quantized geometry information by quantizing the divisional geometry information using the quantization value (quantization parameter) generated by quantization value calculator **5321**.

(593) The plurality of attribute information encoders **5303** generate a plurality of pieces of encoded attribute information by encoding a plurality of pieces of divisional attribute information. For example, the plurality of attribute information encoders **5303** process a plurality of pieces of divisional attribute information in parallel.

(594) As illustrated in FIG. **73**, attribute information encoder **5303** includes quantization value calculator **5331** and entropy encoder **5332**. Quantization value calculator **5331** generates a quantization value (quantization parameter) of divisional attribute information to be encoded. Entropy encoder **5332** calculates quantized attribute information by quantizing the divisional attribute information using the quantization value (quantization parameter) generated by quantization value calculator **5331**.

(595) Additional information encoder **5304** generates encoded additional information by encoding additional information included in the point cloud data and additional information concerning the data division generated in the division by divider **5301**.

(596) Multiplexer **5305** generates encoded data (encoded stream) by multiplexing a plurality of pieces of encoded geometry information, a plurality of pieces of encoded attribute information, and encoded additional information, and transmits the generated encoded data. The encoded additional information is used for decoding.

(597) Note that, although FIG. 71 shows an example in which there are two geometry information encoders 5302 and two attribute information encoders 5303, the number of geometry information encoders 5302 and the number of attribute information encoders 5303 may be one, or three or more. The plurality of pieces of divisional data may be processed in parallel in the same chip, such as by a plurality of cores of a CPU, processed in parallel by cores of a plurality of chips, or processed in parallel by a plurality of cores of a plurality of chips.

(598) Next, a decoding process will be described. FIG. 74 is a block diagram illustrating a configuration of first decoder 5340. FIG. 75 is a block diagram illustrating a configuration of geometry information decoder 5342 and attribute information decoder 5343.

(599) First decoder 5340 reproduces point cloud data by decoding encoded data (encoded stream) generated by encoding the point cloud data in the first encoding method (GPCC). First decoder 5340 includes demultiplexer 5341, a plurality of geometry information decoders 5342, a plurality of attribute information decoders 5343, additional information decoder 5344, and combiner 5345.

(600) Demultiplexer 5341 generates a plurality of pieces of encoded geometry information, a plurality of pieces of encoded attribute information, and encoded additional information by demultiplexing encoded data (encoded stream).

(601) The plurality of geometry information decoders 5342 generate a plurality of pieces of quantized geometry information by decoding a plurality of pieces of encoded geometry information. For example, the plurality of geometry information decoders 5342 process a plurality of pieces of encoded geometry information in parallel.

(602) As illustrated in FIG. 75, geometry information decoder 5342 includes quantization value calculator 5351 and entropy decoder 5352. Quantization value calculator 5351 generates a quantization value of quantized geometry information. Entropy decoder 5352 calculates geometry information by inverse-quantizing the quantized geometry information using the quantization value generated by quantization value calculator 5351.

(603) The plurality of attribute information decoders 5343 generate a plurality of pieces of divisional attribute information by decoding a plurality of pieces of encoded attribute information. For example, the plurality of attribute information decoders 5343 process a plurality of pieces of encoded attribute information in parallel.

(604) As illustrated in FIG. 75, attribute information decoder 5343 includes quantization value calculator 5361 and entropy decoder 5362. Quantization value calculator 5361 generates a quantization value of quantized attribute information. Entropy decoder 5362 calculates attribute information by inverse-quantizing the quantized attribute information using the quantization value generated by quantization value calculator 5361.

(605) The plurality of additional information decoders 5344 generate additional information by decoding encoded additional information.

(606) Combiner 5345 generates geometry information by combining a plurality of pieces of divisional geometry information using additional information. Combiner 5345 generates attribute information by combining a plurality of pieces of divisional attribute information using additional information. For example, combiner 5345 first generates point cloud data associated with a tile by combining decoded point cloud data associated with slices using slice additional information. Combiner 5345 then reproduces the original point cloud data by combining point cloud data associated with tiles using tile additional information.

(607) Note that, although FIG. 74 shows an example in which there are two geometry information decoders 5342 and two attribute information decoders 5343, the number of geometry information decoders 5342 and the number of attribute information decoders 5343 may be one, or three or more. The plurality of pieces of divisional data may be processed in parallel in the same chip, such as by a plurality of cores of a CPU, processed in parallel by cores of a plurality of chips, or processed in parallel by a plurality of cores of a plurality of chips.

(608) [Method of Determining Quantization Parameter]

(609) FIG. 76 is a flowchart illustrating an example of a process concerning determination of a quantization value (quantization parameter value: QP value) in the encoding of geometry information (geometry) or the encoding of attribute information (attribute).

(610) A QP value is determined by considering the coding efficiency on a basis of data units of geometry information or attribute information forming a PCC frame, for example. When the data unit is a tile or slice resulting from division, the QP value is determined on a basis of divisional data units by considering the coding efficiency of the divisional data units. The QP value may be determined on a basis of data units before division.

(611) As illustrated in FIG. 76, the three-dimensional data encoding device determines a QP value used for the encoding of geometry information (S5301). The three-dimensional data encoding device may determine the QP value for each of a plurality of divisional slices in a predetermined manner. Specifically, the three-dimensional data encoding device determines the QP value based on the characteristics or quality of the data of the geometry information. For example, the three-dimensional data encoding device may determine the density of point cloud data for each data unit, that is, the number of points per unit area belonging to each slice, and determine a value corresponding to the density of point cloud data as the QP value. Alternatively, the three-dimensional data encoding device may determine, as the QP value, any of the following values corresponding to geometry information: the number of points of point cloud data, the distribution of points of point cloud data, the imbalance of points of point cloud data, a feature quantity obtained from information on points, the number of feature points, or a recognized object. The three-dimensional data encoding device may also determine an object associated with geometry information of a map and determine the QP value based on the object based on the geometry information, or may determine the QP value based on information or a feature quantity obtained by projecting three-dimensional point cloud onto a two-dimensional plane. The corresponding QP value may be stored in a memory in advance in the form of a table that associates the QP value with the density, the number of points, the distribution of points, or the imbalance of points of point cloud data. The corresponding QP value may also be stored in a memory in advance in the form of a table that associates the QP value with a feature quantity or the number of feature points obtained from information on points or an object recognized based on the information on points. The corresponding QP value may be determined based on a result of simulation of the coding efficiency or the like using various QP values in the encoding of the geometry information concerning point cloud data.

(612) The three-dimensional data encoding device then determines a reference value (default value) of and difference information (quantization delta) on the QP value for geometry information (S5302). Specifically, the three-dimensional data encoding device determines a reference value and difference information to be transmitted using the determined QP value in a predetermined manner, and sets (adds) the determined reference value and difference information in at least one of the additional information or the header of the data.

(613) The three-dimensional data encoding device then determines a QP value used for the encoding of attribute information (S5303). The three-dimensional data encoding device may determine the QP value for each of a plurality of divisional slices in a predetermined manner. Specifically, the three-dimensional data encoding device determines the QP value based on the characteristics or quality of the data of the attribute information. For example, the three-dimensional data encoding device may determine the QP value on a basis of data units based on the characteristics of the attribute information. Color characteristics include luminance, chromaticity, and chroma, a histogram thereof, and color continuity, for example. When the attribute information is reflectance, the QP value may be determined based on information based on the reflectance. For example, when a face is detected as an object from point cloud data, the three-dimensional data encoding device may determine a high-quality QP value for the point cloud data forming the object detected as a face. In this way, the three-dimensional data encoding device may determine the QP

value for the point cloud data forming an object depending on the type of the object.

(614) When a three-dimensional point has a plurality of pieces of attribute information, the three-dimensional data encoding device may determine a different QP value for each piece of attribute information based on the piece of attribute information. Alternatively, the three-dimensional data encoding device may determine a QP value for the plurality of pieces of attribute information based on any one of the pieces of attribute information, or determine a QP value for the plurality of pieces of attribute information based on a plurality of pieces of attribute information.

(615) The three-dimensional data encoding device then determines a reference value (default value) of and difference information (quantization delta) on the QP value for attribute information (S5304). Specifically, the three-dimensional data encoding device determines a reference value and difference information to be transmitted using the determined QP value in a predetermined manner, and sets (adds) the determined reference value and difference information in at least one of the additional information or the header of the data.

(616) The three-dimensional data encoding device then quantizes and encodes the geometry information and the attribute information based on the determined QP values for geometry information and attribute information, respectively (S5305).

(617) Note that although an example has been described in which the QP value for geometry information is determined based on the geometry information, and the QP value for attribute information is determined based on the attribute information, the present disclosure is not limited thereto. For example, the QP values for geometry information and attribute information may be determined based on the geometry information, based on the attribute information, or based on the geometry information and the attribute information.

(618) Note that the QP values for geometry information and attribute information may be adjusted by considering the balance between the quality of the geometry information and the quality of the attribute information in the point cloud data. For example, the QP values for geometry information and attribute information may be set in such a manner that the quality of the geometry information is high, and the quality of the attribute information is lower than the quality of the geometry information. For example, the QP value for attribute information may be determined under a restriction that the QP value for attribute information is equal to or higher than the QP value for geometry information.

(619) The QP value may be adjusted so that encoded data is generated within a predetermined range of rate. For example, when the code amount of the encoding of the preceding data unit is expected to exceed a predetermined rate, that is, when the difference from a predetermined rate is less than a first difference, the QP value may be adjusted to decrease the coding quality so that the difference between the predetermined rate and the code amount of the data unit is less than the first difference. On the other hand, when the difference from the predetermined rate is greater than a second difference, which is greater than the first difference, and there is a substantial difference, the QP value may be adjusted to improve the coding quality of the data unit. The adjustment between data units may be made between PCC frames or between tiles or slices. The adjustment of the QP value for attribute information may be made based on the rate of encoding of geometry information.

(620) Note that, in the flowchart of FIG. 76, the processing concerning geometry information and the processing concerning attribute information may be performed in reverse order or in parallel.

(621) Note that, although the flowchart of FIG. 76 shows a slice-based process as an example, a tile-based process or a process on a basis of other data units can be performed in the same manner as the slice-based process. That is, slice in the flowchart of FIG. 76 can be replaced with tile or other data units.

(622) FIG. 77 is a flowchart illustrating an example of a process of decoding geometry information and attribute information.

(623) As illustrated in FIG. 77, the three-dimensional data decoding device obtains a reference

value and difference information that indicate a QP value for geometry information, and a reference value and difference information that indicate a QP value for attribute information (S5311). Specifically, the three-dimensional data decoding device analyzes one or both of the transmitted metadata or the header of the transmitted encoded data, and obtains reference values and difference information for deriving the QP values.

(624) The three-dimensional data decoding device then derives the QP values using the obtained reference values and difference information in a predetermined manner.

(625) The three-dimensional data decoding device then obtains quantized geometry information, and obtains geometry information by inverse-quantizing the quantized geometry information using the derived QP value (S5313).

(626) The three-dimensional data decoding device then obtains quantized attribute information, and obtains attribute information by inverse-quantizing the quantized attribute information using the derived QP value (S5314).

(627) Next, a method of transmitting a quantization parameter will be described.

(628) FIG. 78 is a diagram for describing a first example of the method of transmitting a quantization parameter. Part (a) of FIG. 78 shows an example of a relationship between QP values.

(629) In FIG. 78, Q.sub.G and Q.sub.A denote an absolute value of a QP value used for the encoding of geometry information and an absolute value of a QP value used for the encoding of attribute information, respectively. Q.sub.G is an example of a first quantization parameter used for quantizing geometry information on each of a plurality of three-dimensional points. $\Delta(Q.sub.A, Q.sub.G)$ denotes difference information that indicates a difference between Q.sub.A and Q.sub.G used for deriving Q.sub.A. That is, Q.sub.A is derived using Q.sub.G and $\Delta(Q.sub.A, Q.sub.G)$. In this way, a QP value is separated into a reference value (absolute value) and difference information (relative value) for transmission. In the decoding, a desired QP value is derived from the transmitted reference value and difference information.

(630) For example, in part (a) of FIG. 78, the absolute value Q.sub.G and the difference information $\Delta(Q.sub.A, Q.sub.G)$ are transmitted, and in the decoding, Q.sub.A is derived by adding $\Delta(Q.sub.A, Q.sub.G)$ to Q.sub.G as shown by Equation G5 below.

$$Q.sub.A = Q.sub.G + \Delta(Q.sub.A, Q.sub.G) \quad (\text{Equation G5})$$

(631) With reference to parts (b) and (c) of FIG. 78, a method of transmitting QP values in a case where point cloud data including geometry information and attribute information is divided into slices will be described. Part (b) of FIG. 78 shows a first example of a relationship between a reference value and difference information for each QP value. Part (c) of FIG. 78 shows a first example of an order of transmission of QP values, geometry information, and attribute information.

(632) For each piece of geometry information and each piece of attribute information, QP values are classified into QP values (frame QP in units of PCC frames and QP values (data QP in units of data units. The QP value used for the encoding determined in step S5301 in FIG. 76 is a QP value in units of data units.

(633) Here, Q.sub.G, which is a QP value used for the encoding of geometry information in units of PCC frames, is used as a reference value, and a QP value in units of data units is generated and transmitted as difference information that indicates the difference from Q.sub.G.

(634) Q.sub.G: a QP value for the encoding of geometry information for a PCC frame, which is transmitted as a reference value “1.” using GPS.

(635) Q.sub.A: a QP value for the encoding of attribute information for a PCC frame, which is transmitted as difference information “2.” using APS.

(636) Q.sub.Gs1, Q.sub.Gs2: QP values for the encoding of geometry information of slice data, which are transmitted as difference information “3.” and “5.” indicating a difference from Q.sub.G, respectively, using the header of the encoded data of the geometry information.

(637) Q.sub.As1, Q.sub.As2: QP values for the encoding of attribute information of slice data, which are transmitted as difference information “4.” and “6.” indicating a difference from Q.sub.A,

respectively, using the header of the encoded data of the attribute information.

(638) Note that information used for deriving a frame QP is described in metadata (GPS, APS) associated with the frame, and information used for deriving a data QP is described in metadata (header of encoded data) associated with the data.

(639) In this way, the data QP is generated and transmitted as difference information indicating a difference from the frame QP. Therefore, the data amount of the data QP can be reduced.

(640) In each piece of encoded data, first decoder 5340 refers to metadata indicated by an arrow in part (c) of FIG. 78, and obtains a reference value and difference information associated with the encoded data. First decoder 5340 then derives a QP value corresponding to the encoded data to be decoded based on the obtained reference value and difference information.

(641) For example, first decoder 5340 obtains the reference information “1.” and the difference information “2.” and “6.” indicated by arrows in part (c) of FIG. 78 from the metadata or the header, and derives the QP value of As.sub.2 by adding the difference information “2.” and “6.” to the reference information “1.” as shown by Equation G6 below.

$$Q.\text{sub.As2} = Q.\text{sub.G} + \Delta(Q.\text{sub.A}, Q.\text{sub.G}) + \Delta(Q.\text{sub.As2}, Q.\text{sub.A}) \quad (\text{Equation G6})$$

(642) Point cloud data includes geometry information and zero or more pieces of attribute information. That is, point cloud data may include no attribute information or a plurality of pieces of attribute information.

(643) For example, one three-dimensional point may have, as attribute information, color information, color information and reflectance information, or one or more pieces of color information linked to one or more pieces of point-of-view information.

(644) Here, an example of a case where one three-dimensional point has two pieces of color information and reflectance information will be described with reference to FIG. 79. FIG. 79 is a diagram for describing a second example of the method of transmitting a quantization parameter. Part (a) of FIG. 79 is a diagram illustrating a second example of the relationship between a reference value and difference information for each QP value. Part (b) of FIG. 79 is a diagram illustrating a second example of the order of transmission of QP values, geometry information, and attribute information.

(645) Q.sub.G is an example of the first quantization parameter as in FIG. 78.

(646) Two pieces of color information are indicated by luminance (luma) Y and chrominances (chroma) Cb, Cr, respectively. Q.sub.Y1, which is a QP value used for the encoding of luminance Y1 of a first color, is derived from Q.sub.G, which is a reference value, and $\Delta(Q.\text{sub.Y1}, Q.\text{sub.G})$, which indicates the difference between Q.sub.Y1 and Q.sub.G. Luminance Y1 is an example of a first luminance, and Q.sub.Y1 is an example of a second quantization parameter used for quantizing luminance Y1 as the first luminance. $\Delta(Q.\text{sub.Y1}, Q.\text{sub.G})$ is difference information “2.”

(647) Q.sub.Cb1 and Q.sub.Cr1, which are QP values used for the encoding of chrominances Cb1 and Cr1 of the first color, are derived from Q.sub.Y1 and $\Delta(Q.\text{sub.Cb1}, Q.\text{sub.Y1})$ and $\Delta(Q.\text{sub.Cr1}, Q.\text{sub.Y1})$, which indicate the difference between Q.sub.Cb1 and Q.sub.Y1 and the difference between Q.sub.Cr1 and Q.sub.Y1, respectively. Chrominances Cb1 and Cr1 are examples of a first chrominance, and Q.sub.Cb1 and Q.sub.Cr1 are examples of a third quantization parameter used for quantizing chrominances Cb1 and Cr1 as the first chrominance. $\Delta(Q.\text{sub.Cb1}, Q.\text{sub.Y1})$ is difference information “3.”, and $\Delta(Q.\text{sub.Cr1}, Q.\text{sub.Y1})$ is difference information “4.”. $\Delta(Q.\text{sub.Cb1}, Q.\text{sub.Y1})$ and $\Delta(Q.\text{sub.Cr1}, Q.\text{sub.Y1})$ are examples of a first difference.

(648) Note that Q.sub.Cb1 and Q.sub.Cr1 may be identical values or a common value. When a common value is used, one of Q.sub.Cb1 and Q.sub.Cr1 can be used, and the other can be omitted.

(649) Q.sub.Y1D, which is a QP value used for the encoding of luminance Y1D of the first color in the slice data, is derived from Q.sub.Y1 and $\Delta(Q.\text{sub.Y1D}, Q.\text{sub.Y1})$ indicating the difference between Q.sub.Y1D and Q.sub.Y1. Luminance Y1D of the first color in the slice data is an

example of the first luminance of one or more three-dimensional points included in the subspace, and Q.sub.Y1D is an example of a fifth quantization parameter used for quantizing luminance Y.sub.1D. $\Delta(Q.sub.Y1D, Q.sub.Y1)$ is difference information “10.”, and an example of a second difference.

(650) Similarly, Q.sub.Cb1D and Q.sub.Cr1D, which are QP values used for the encoding of chrominances Cb1D and Cr1D of the first color in the slice data, are derived from Q.sub.Cb1 and $\Delta(Q.sub.Cb1D, Q.sub.Cb1)$ indicating the difference between Q.sub.Cb1D and Q.sub.Cb1 and Q.sub.Cr1 and $\Delta(Q.sub.Cr1D, Q.sub.Cr1)$ indicating the difference between Q.sub.Cr1D and Q.sub.Cr1, respectively. Chrominances Cb1D and Cr1D of the first color in the slice data are examples of the first chrominance of one or more three-dimensional points included in the subspace, and Q.sub.Gb1D and Q.sub.Cr1D are examples of a sixth quantization parameter used for quantizing chrominances Cb1D and Cr1D. $\Delta(Q.sub.Cb1D, Q.sub.Cb1)$ is difference information “11.”, and $\Delta(Q.sub.Cr1D, Q.sub.Cr1)$ is difference information “12.”. $\Delta(Q.sub.Cb1D, Q.sub.Cb1)$ and $\Delta(Q.sub.Cr1D, Q.sub.Cr1)$ are examples of a third difference.

(651) The relationship between QP values for the first color holds for a second color, so that redundant descriptions will be omitted.

(652) Q.sub.R, which is a QP value used for the encoding of reflectance R, is derived from Q.sub.G, which is a reference value, and $\Delta(Q.sub.R, Q.sub.G)$, which indicates the difference between Q.sub.R and Q.sub.G. Q.sub.R is an example of a fourth quantization parameter used for quantizing reflectance R. $\Delta(Q.sub.R, Q.sub.G)$ is difference information “8.”.

(653) Q.sub.RD, which is a QP value used for the encoding of reflectance RD in the slice data, is derived from Q.sub.R and $\Delta(Q.sub.RD, Q.sub.R)$, which indicates the difference between Q.sub.RD and Q.sub.R. $\Delta(Q.sub.RD, Q.sub.R)$ is difference information “16.”.

(654) As described above, difference information “9.” to “16.” indicates the difference between a data QP and a frame QP.

(655) Note that when the values of the data QP and the frame QP are the same, for example, the difference information may be set at 0, or may not be transmitted, and the absence of the transmission may be regarded as difference information of 0.

(656) When obtaining chrominance Cr2 of the second color by decoding, for example, first decoder **5340** obtains reference information “1.” and difference information “5.”, “7.”, and “15.” indicated by arrows in part (b) of FIG. **79** from the metadata or the header, and derives the QP value of chrominance Cr2 by adding difference information “5.”, “7.”, and “15.” to reference information “1.” as shown by Equation G7 below.

$$Q.sub.Cr2D = Q.sub.G + \Delta(Q.sub.Y2, Q.sub.G) + \Delta(Q.sub.Cr2, Q.sub.Y2) + \Delta(Q.sub.Cr2D, Q.sub.Cr2)$$

(Equation G7)

(657) Next, an example of a case where geometry information and attribute information are divided into two tiles and then divided into two slices will be described with reference to FIG. **80**. FIG. **80** is a diagram for describing a third example of the method of transmitting a quantization parameter. Part (a) of FIG. **80** shows a third example of the relationship between a reference value and difference information for each QP value. Part (b) of FIG. **80** shows a third example of the order of transmission of QP values, geometry information, and attribute information. Part (c) of FIG. **80** describes an intermediate generated value for difference information in the third example.

(658) When geometry information and attribute information are divided into a plurality of tiles and then further divided into a plurality of slices, as illustrated in part (c) of FIG. **80**, after the attribute information is divided into tiles, a QP value (Q.sub.At1) and difference information $\Delta(Q.sub.At1, Q.sub.A)$ for each tile are generated as intermediate generated values. After the tile is divided into slices, QP values (Q.sub.At1s1, Q.sub.At1s2) and difference information ($\Delta(Q.sub.At1s1, Q.sub.At1)$, $\Delta(Q.sub.At1s2, Q.sub.At1)$) are generated for each slice.

(659) In this case, difference information “4.” in part (a) of FIG. **80** is derived according to Equation G8 below.

$$\Delta(Q.sub.At1s1,Q.sub.A)=\Delta(Q.sub.At1,Q.sub.A)+\Delta(Q.sub.At1s1,Q.sub.At1) \quad (\text{Equation G8})$$

(660) When obtaining attribute information A.sub.t2s1 for slice 1 in tile 2 by decoding, for example, first decoder **5340** obtains reference information “1.” and difference information “2.” and “8.” indicated by arrows in part (b) of FIG. **80** from the metadata or the header, and derives the QP value of attribute information A.sub.t2s1 by adding difference information “2.” and “8.” to reference information “1.” as shown by Equation G9 below.

$$Q.sub.At2s1=Q.sub.G+\Delta(Q.sub.At2s1,Q.sub.A)+\Delta(Q.sub.A,Q.sub.G) \quad (\text{Equation G9})$$

(661) Next, a flow of a process of encoding point cloud data and a flow of a process of decoding point cloud data according to this embodiment will be described. FIG. **81** is a flowchart of a process of encoding point cloud data according to this embodiment.

(662) First, the three-dimensional data encoding device determines a division method to be used (**S5321**). The division method includes a determination of whether to perform tile division or not and a determination of whether to perform slice division or not. The division method may include the number of tiles or slices in the case where tile division or slice division is performed, and the type of division, for example. The type of division is a scheme based on an object shape, a scheme based on map information or geometry information, or a scheme based on a data amount or processing amount, for example. Note that the division method may be determined in advance.

(663) When tile division is to be performed (if Yes in **S5322**), the three-dimensional data encoding device generates a plurality of pieces of tile geometry information and a plurality of pieces of tile attribute information by dividing the geometry information and the attribute information on a tile basis (**S5323**). The three-dimensional data encoding device also generates tile additional information concerning the tile division.

(664) When slice division is to be performed (if Yes in **S5324**), the three-dimensional data encoding device generates a plurality of pieces of divisional geometry information and a plurality of pieces of divisional attribute information by dividing the plurality of pieces of tile geometry information and the plurality of pieces of tile attribute information (or the geometry information and the attribute information) (**S5325**). The three-dimensional data encoding device also generates geometry slice additional information and attribute slice additional information concerning the slice division.

(665) The three-dimensional data encoding device then generates a plurality of pieces of encoded geometry information and a plurality of pieces of encoded attribute information by encoding each of the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information (**S5326**). The three-dimensional data encoding device also generates dependency information.

(666) The three-dimensional data encoding device then generates encoded data (encoded stream) by integrating (multiplexing) the plurality of pieces of encoded geometry information, the plurality of pieces of encoded attribute information and the additional information into a NAL unit (**S5327**). The three-dimensional data encoding device also transmits the generated encoded data.

(667) FIG. **82** is a flowchart illustrating an example of a process of determining a QP value and updating additional information in the tile division (**S5323**) and the slice division (**S5325**).

(668) In steps **S5323** and **S5325**, tile geometry information and tile attribute information and/or slice geometry information and slice attribute information may be independently divided in respective manners, or may be collectively divided in a common manner. In this way, additional information divided on a tile basis and/or on a slice basis is generated.

(669) In these steps, the three-dimensional data encoding device determines a reference value and difference information for a QP value on a divisional tile basis and/or on a divisional slice basis (**S5331**). Specifically, the three-dimensional data encoding device determines reference value and difference information such as those illustrated in FIGS. **78** to **80**.

(670) The three-dimensional data encoding device then updates the additional information to include the determined reference value and difference information (**S5332**).

(671) FIG. **83** is a flowchart illustrating an example of a process in encoding (S5326).

(672) The three-dimensional data encoding device encodes each of the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information (S5341). Specifically, the three-dimensional data encoding device encodes each of the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information using the determined QP value.

(673) The three-dimensional data encoding device then continues the encoding process until a condition for stopping the encoding process is satisfied, such as until there is no data to be encoded (S5342).

(674) FIG. **84** is a flowchart illustrating a process of decoding point cloud data according to this embodiment. First, the three-dimensional data decoding device determines the division method by analyzing additional information (tile additional information, geometry slice additional information, and attribute slice additional information) concerning the division method included in encoded data (encoded stream) (S5351). The division method includes a determination of whether to perform tile division or not and a determination of whether to perform slice division or not. The division method may include the number of tiles or slices in the case where tile division or slice division is performed, and the type of division, for example.

(675) The three-dimensional data decoding device then generates divisional geometry information and divisional attribute information by decoding a plurality of pieces of encoded geometry information and a plurality of pieces of encoded attribute information included in the encoded data using dependency information included in the encoded data (S5352).

(676) If the additional information indicates that slice division has been performed (if Yes in S5353), the three-dimensional data decoding device generates a plurality of pieces of tile geometry information and a plurality of pieces of tile attribute information by combining the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information based on the geometry slice additional information and the attribute slice additional information (S5354).

(677) If the additional information indicates that tile division has been performed (if Yes in S5355), the three-dimensional data decoding device generates geometry information and attribute information by combining the plurality of pieces of tile geometry information and the plurality of pieces of tile attribute information (the plurality of pieces of divisional geometry information and the plurality of pieces of divisional attribute information) based on the tile additional information (S5356).

(678) FIG. **85** is a flowchart illustrating an example of a process of obtaining QP values and decoding a QP value for a slice or tile in the combining of information divided into slices (S5354) and the combining of information divided into tiles (S5356).

(679) Pieces of slice geometry information and pieces of slice attribute information or pieces of tile geometry information or pieces of tile attribute information may be combined in respective manners or in the same manner.

(680) The three-dimensional data decoding device obtains the reference value and the difference information by decoding the additional information in the encoded stream (S5361).

(681) The three-dimensional data decoding device then calculates a quantization value using the decoded reference value and difference information, and updates the QP value used for inverse quantization to the calculated QP value (S5362). In this way, a QP value for inverse quantization of quantized attribute information for each tile or slice can be derived.

(682) The three-dimensional data decoding device then continues the decoding process until a condition for stopping the decoding process is satisfied, such as until there is no data to be decoded (S5363).

(683) FIG. **86** is a diagram illustrating a syntax example of GPS. FIG. **87** is a diagram illustrating a syntax example of APS. FIG. **88** is a diagram illustrating a syntax example of a header of geometry

information. FIG. **89** is a diagram illustrating a syntax example of a header of attribute information. (684) As illustrated in FIG. **86**, for example, GPS, which is additional information of geometry information, includes QP_value, which indicates an absolute value used as a reference for deriving a QP value. QP_value corresponds to Q.sub.G illustrated in FIGS. **78** to **80**.

(685) As illustrated in FIG. **87**, for example, when a three-dimensional point has a plurality of pieces of color information associated with a plurality of points of view, APS, which is additional information of attribute information, may define a default point of view, and a 0-th piece of attribute information may always describe information on the default point of view. For example, when decoding or displaying a single piece of color information, the three-dimensional data encoding device can decode or display the 0-th piece of attribute information.

(686) APS includes QP_delta_Attribute_to_Geometry. QP_delta_Attribute_to_Geometry is difference information indicating the difference from the reference value (QP_value) described in GPS. The difference information indicates a difference in luminance when the attribute information is color information, for example.

(687) GPS may include a flag that indicates whether or not Geometry_header (header of the geometry information) includes difference information used for calculating a QP value. APS may include a flag that indicates whether or not Attribute_header (header of the attribute information) includes difference information used for calculating a QP value. The flag may indicate whether or not the attribute information includes difference information indicating the difference of a data QP from a frame QP, which is used for calculating the data QP.

(688) When a first color of attribute information is indicated by a first luminance and a first chrominance, in the quantization of the first luminance using a second quantization parameter and the quantization of the first chrominance using a third quantization parameter, if the quantizations are performed using a fifth quantization parameter and a sixth quantization parameter, the encoded stream may include identification information (flag) that indicates that the quantizations are performed using the fifth quantization parameter and the sixth quantization parameter.

(689) As illustrated in FIG. **88**, the header of the geometry information may include QP_delta_data_to_frame, which is difference information indicating the difference from the reference value (QP_value) described in GPS. The header of the geometry information may be divided into pieces of information associated with tiles and/or slices, and a QP value corresponding to each tile and/or slice may be indicated.

(690) As illustrated in FIG. **89**, the header of the attribute information may include QP_delta_data_to_frame, which is difference information indicating the difference from the QP value described in APS.

(691) Although the reference value of a QP value has been described as being a QP value of geometry information for a PCC frame with reference to FIGS. **78** to **80**, the present disclosure is not limited thereto, and other values may be used as a reference value.

(692) FIG. **90** is a diagram for describing another example of the method of transmitting a quantization parameter.

(693) Parts (a) and (b) of FIG. **90** illustrate a fourth example, in which common reference value Q is set based on QP values of geometry information and attribute information for a PCC frame. In the fourth example, reference value Q is stored in GPS, difference information indicating the difference of a QP value (Q.sub.G) of geometry information from reference value Q is stored in GPS, and difference information indicating the differences of QP values (Q.sub.Y and Q.sub.R) of attribute information from reference value Q is stored in APS. Note that reference value Q may be stored in SPS.

(694) Parts (c) and (d) of FIG. **90** illustrate a fifth example, in which a different reference value is set for each of geometry information and attribute information. In the fifth example, reference QP values (absolute values) of geometry information and attribute information are stored in GPS and APS, respectively. That is, reference value Q.sub.G is set in geometry information, reference value

Q.sub.Y is set in color information of attribute information, and reference value Q.sub.R is set as reflectance of attribute information. In this way, a reference value of a QP value may be set for each of geometry information and a plurality of kinds of attribute information. Note that the fifth example may be combined with another example. That is, Q.sub.A in the first example, or Q.sub.Y1, Q.sub.Y2, and Q.sub.R in the second example may be a reference value of a QP value. (695) Parts (e) and (f) of FIG. 90 illustrate a sixth example, in which when there is a plurality of PCC frames, a common reference value Q is set for the plurality of PCC frames. In the sixth example, reference value Q is stored in SPS or GPS, and difference information indicating the difference between the QP value of the geometry information and the reference value for each PCC frame is stored in GPS. Note that, within the range of a random access unit, such as GOF, for example, the leading frame of the random access unit may be designated as a reference value, and difference information $\Delta(Q.sub.G(1), Q.sub.G(0))$ indicating the differences between the PCC frames may be transmitted.

(696) Note that, even when a tile or a slice is further divided, difference information indicating the difference from the QP value of the unit of division is stored in the data header and transmitted in the same manner.

(697) FIG. 91 is a diagram for describing another example of the method of transmitting a quantization parameter.

(698) Parts (a) and (b) of FIG. 91 illustrate a seventh example, in which common reference value Q.sub.G is set for geometry information and attribute information of a PCC frame. In the seventh example, reference value Q.sub.G is stored in GPS, and difference information indicating the differences from the geometry information or the attribute information is stored in the respective data headers. Reference value Q.sub.G may be stored in SPS.

(699) Parts (c) and (d) of FIG. 91 shows an eighth example, in which a QP value of attribute information is indicated by difference information indicating the difference from a QP value of geometry information belonging to the same slice and tile. In the eighth example, reference value Q.sub.G may be stored in SPS.

(700) FIG. 92 is a diagram for describing a ninth example of the method of transmitting a quantization parameter.

(701) Parts (a) and (b) illustrate the ninth example, in which a plurality of pieces of attribute information has a common QP value, and each piece of attribute information indicates difference information indicating the difference between the common QP value and the QP value of geometry information and difference information indicating the difference from the value of the attribute information and the common QP value.

(702) FIG. 93 is a diagram for describing an example of control of a QP value.

(703) As the value of the quantization parameter decreases, the quality improves, while the coding efficiency decreases because more bits are required.

(704) For example, when encoding three-dimensional point cloud data by dividing the three-dimensional point cloud data into tiles, if point cloud data included in a tile is a main road, the point cloud data is encoded using a previously defined QP value of attribute information. On the other hand, peripheral tiles do not include important information, and therefore, the coding efficiency may be able to be improved by setting the difference information of the QP value at a positive value to reduce the quality of the data.

(705) Furthermore, when encoding the three-dimensional point cloud data divided into tiles by dividing the tiles into slices, a sidewalk, a tree, and a building are important for positional estimation (localization and mapping) in automatic driving, so that the QP value is set at a negative value. On the other hand, a moving body and other objects are less important, so that the QP value is set at a positive value.

(706) Part (b) of FIG. 93 shows an example in which difference information is derived in a case where a quantization delta value is set in advance based on the object included in a tile or slice. For

example, when divisional data is slice data on a “building” included in a tile of a “main road”, the difference information is -5 , which is derived by summing the quantization delta value of 0 of the tile of a “main road” and the quantization delta value of -5 of the slice data on a “building”.

(707) FIG. **94** is a flowchart illustrating an example of a method of determining a QP value based on the quality of an object.

(708) The three-dimensional data encoding device divides point cloud data into one or more tiles based on map information, and determines an object included in the one or more tiles (S5371).

Specifically, the three-dimensional data encoding device performs an object recognition process for recognizing what an object is using a leaning model obtained by machine learning, for example.

(709) The three-dimensional data encoding device then determines whether to encode a tile to be processed with high quality or not (S5372). To encode with high quality means encoding at a bitrate higher than a predetermined rate, for example.

(710) When the tile to be processed is to be encoded with high quality (if Yes in S5372), the three-dimensional data encoding device then sets the QP value of the tile so that the coding efficiency is high (S5373).

(711) On the other hand, when the tile to be processed is not to be encoded with high quality (if No in S5372), the three-dimensional data encoding device sets the QP value of the tile so that the coding efficiency is low (S5374).

(712) Following step S5373 or S5374, the three-dimensional data encoding device determines the object in the tile, and divides the tile into one or more slices (S5375).

(713) The three-dimensional data encoding device then determines whether to encode a slice to be processed with high quality or not (S5376).

(714) When the slice to be processed is to be encoded with high quality (if Yes in S5376), the three-dimensional data encoding device then sets the QP value of the slice so that the coding efficiency is high (S5377).

(715) On the other hand, when the slice to be processed is not to be encoded with high quality (if No in S5376), the three-dimensional data encoding device sets the QP value of the slice so that the coding efficiency is low (S5378).

(716) The three-dimensional data encoding device then determines a reference value and difference information to be transmitted based on the set QP value in a predetermined manner, and stores the determined reference value and difference information in at least one of the additional information and the header of the data (S5379).

(717) The three-dimensional data encoding device then quantizes and encodes the geometry information and the attribute information based on the determined QP value (S5380).

(718) FIG. **95** is a flowchart illustrating an example of a method of determining a QP value based on a rate control.

(719) The three-dimensional data encoding device sequentially encodes point cloud data (S5381).

(720) The three-dimensional data encoding device then determines a rate control status concerning the encoding process from the code amount of the encoded data and the occupancy of an encoding buffer, and determines the quality of the subsequent encoding (S5382).

(721) The three-dimensional data encoding device then determines whether or not to increase the encoding quality (S5383).

(722) When the encoding quality is to be increased (if Yes in S5383), the three-dimensional data encoding device sets the QP value of the tile so that the encoding quality is higher (S5384).

(723) On the other hand, when the encoding quality is not to be increased (if No in S5383), the three-dimensional data encoding device sets the QP value of the tile so that the encoding quality is lower (S5385).

(724) The three-dimensional data encoding device then determines a reference value and difference information to be transmitted based on the set QP value in a predetermined manner, and stores the determined reference value and difference information in at least one of the additional information

and the header of the data (S5386).

(725) The three-dimensional data encoding device then quantizes and encodes the geometry information and the attribute information based on the determined QP value (S5387).

(726) As described above, the three-dimensional data encoding device according to this embodiment performs the process illustrated in FIG. 96. First, the three-dimensional data encoding device quantizes geometry information on each of a plurality of three-dimensional points using a first quantization parameter (S5391). The three-dimensional data encoding device quantizes a first luminance using a second quantization parameter and quantizes a first chrominance using a third quantization parameter, the first luminance and the first chrominance indicating a first color among attribute information on each of the plurality of three-dimensional points (S5392). The three-dimensional data encoding device generates a bitstream including the quantized geometry information, the quantized first luminance, the quantized first chrominance, the first quantization parameter, the second quantization parameter, and a first difference between the second quantization parameter and the third quantization parameter (S5393).

(727) With such a configuration, since the third quantization parameter is indicated by the first difference from the second quantization parameter in the bitstream, the coding efficiency can be improved.

(728) For example, the three-dimensional data encoding device further quantizes a reflectance among the attribute information on each of the plurality of three-dimensional points using a fourth quantization parameter. Furthermore, in the generation described above, the bitstream generated further includes the quantized reflectance and the fourth quantization parameter.

(729) For example, in the quantization using the second quantization parameter, for each of a plurality of subspaces obtained by dividing a current space including the plurality of three-dimensional points, the first luminance of one or more three-dimensional points included in the subspace is quantized further using a fifth quantization parameter. In the quantization using the third quantization parameter, the first chrominance of the one or more three-dimensional points is quantized further using a sixth quantization parameter. In the generation described above, the bitstream generated further includes a second difference between the second quantization parameter and the fifth quantization parameter and a third difference between the third quantization parameter and the sixth quantization parameter.

(730) With such a configuration, since the fifth quantization parameter is indicated by the second difference from the second quantization parameter and the sixth quantization parameter is indicated by the third difference from the third quantization parameter in the bitstream, the coding efficiency can be improved.

(731) For example, in the generating described above, the bitstream generated further includes identification information indicating that the fifth quantization parameter and the sixth quantization parameter have been used in the quantization using the second quantization parameter and the quantization using the third quantization parameter, respectively.

(732) With such a configuration, the three-dimensional data decoding device having obtained the bitstream can determine from the identification information that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed, so that the processing load of the decoding process can be reduced.

(733) For example, the three-dimensional data encoding device further quantizes a second luminance using a seventh quantization parameter and quantizes a second chrominance using an eighth quantization parameter, the second luminance and the second chrominance indicating a second color among the attribute information of each of the plurality of three-dimensional points. In the generation described above, the bitstream generated further includes the quantized second luminance, the quantized second chrominance, the seventh quantization parameter, and a fourth difference between the seventh quantization parameter and the eighth quantization parameter.

(734) With such a configuration, since the eighth quantization parameter is indicated by the fourth

difference from the seventh quantization parameter in the bitstream, the coding efficiency can be improved. In addition, two types of color information can be included in the attribute information on a three-dimensional point.

(735) For example, the three-dimensional data encoding device includes a processor and memory, and the processor performs the process described above using the memory.

(736) The three-dimensional data decoding device according to this embodiment performs the process illustrated in FIG. 97. First, the three-dimensional data decoding device obtains quantized geometry information, a quantized first luminance, a quantized first chrominance, a first quantization parameter, a second quantization parameter, and a first difference between the second quantization parameter and a third quantization parameter, by obtaining a bitstream (S5394). The three-dimensional data decoding device calculates geometry information on a plurality of three-dimensional points by inverse-quantizing the quantized geometry information using the first quantization information (S5395). Of a first luminance and a first chrominance indicating a first color of the plurality of three-dimensional points, the three-dimensional data decoding device calculates the first luminance by inverse-quantizing the quantized first luminance using the second quantization parameter (S5396). The three-dimensional data decoding device calculates the first chrominance by inverse-quantizing the quantized first chrominance using the third quantization parameter obtained from the second quantization parameter and the first difference (S5397).

(737) In this way, the three-dimensional data decoding device can correctly decode geometry information and attribute information on a three-dimensional point.

(738) For example, in the obtaining, a quantized reflectance and a fourth quantization parameter are further obtained by obtaining the bitstream. The three-dimensional data decoding device further calculates a reflectance of the plurality of three-dimensional points by inverse-quantizing the quantized reflectance using the fourth quantization parameter.

(739) Therefore, the three-dimensional data decoding device can correctly decode the reflectance of a three-dimensional point.

(740) For example, in the obtaining, a second difference between the second quantization parameter and a fifth quantization parameter and a third difference between the third quantization parameter and a sixth quantization parameter are further obtained by obtaining the bitstream. In the calculating of the first luminance, a first luminance of one or more three-dimensional points is calculated by inverse-quantizing the quantized first luminance using the second quantization parameter and the fifth quantization parameter obtained from the second difference, the one or more three-dimensional points being included in each subspace obtained by dividing a current space including the plurality of three-dimensional points, the quantized first luminance being the luminance obtained by quantizing the first luminance of the one or more three-dimensional points using the second quantization parameter and the fifth quantization parameter. In the calculation of the first chrominance, a first chrominance of the one or more three-dimensional points is calculated by inverse-quantizing the quantized first chrominance using the third quantization parameter and the sixth quantization parameter obtained from the third difference, the quantized first chrominance being the chrominance obtained by quantizing the first chrominance of the at least one three-dimensional point using the third quantization parameter and the sixth quantization parameter.

(741) For example, in the obtaining, identification information indicating that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed is further obtained by obtaining the bitstream. In the calculation of the first luminance, when the identification information indicates that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed, the quantized first luminance is determined to be a luminance obtained by quantizing the first luminance of the one or more three-dimensional points. In the calculation of the first chrominance, when the identification information indicates that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been

performed, the quantized first chrominance is determined to be a chrominance obtained by quantizing the first chrominance of the one or more three-dimensional points.

(742) With such a configuration, the three-dimensional data decoding device can determine from the identification information that the quantization using the fifth quantization parameter and the quantization using the sixth quantization parameter have been performed, so that the processing load of the decoding process can be reduced.

(743) For example, in the obtaining, a quantized second luminance, a quantized second chrominance, a seventh quantization parameter, and a fourth difference between the seventh quantization parameter and an eighth quantization parameter is further obtained by obtaining the bitstream. Of a second luminance and a second chrominance that indicate a second color of the plurality of three-dimensional points, the three-dimensional data decoding device further calculates the second luminance by inverse-quantizing the quantized second luminance using the seventh quantization parameter. The three-dimensional data decoding device further calculates the second chrominance by inverse-quantizing the quantized second chrominance using the eighth quantization parameter obtained from the seventh quantization parameter and the fourth difference.

(744) In this way, the three-dimensional data decoding device can correctly decode the second color of a three-dimensional point.

(745) For example, the three-dimensional data decoding device includes a processor and memory, and the processor performs the process described above using the memory.

(746) A three-dimensional data encoding device, a three-dimensional data decoding device, and the like according to the embodiments of the present disclosure have been described above, but the present disclosure is not limited to these embodiments.

(747) Note that each of the processors included in the three-dimensional data encoding device, the three-dimensional data decoding device, and the like according to the above embodiments is typically implemented as a large-scale integrated (LSI) circuit, which is an integrated circuit (IC). These may take the form of individual chips, or may be partially or entirely packaged into a single chip.

(748) Such IC is not limited to an LSI, and thus may be implemented as a dedicated circuit or a general-purpose processor. Alternatively, a field programmable gate array (FPGA) that allows for programming after the manufacture of an LSI, or a reconfigurable processor that allows for reconfiguration of the connection and the setting of circuit cells inside an LSI may be employed.

(749) Moreover, in the above embodiments, the structural components may be implemented as dedicated hardware or may be realized by executing a software program suited to such structural components. Alternatively, the structural components may be implemented by a program executor such as a CPU or a processor reading out and executing the software program recorded in a recording medium such as a hard disk or a semiconductor memory.

(750) The present disclosure may also be implemented as a three-dimensional data encoding method, a three-dimensional data decoding method, or the like executed by the three-dimensional data encoding device, the three-dimensional data decoding device, and the like.

(751) Also, the divisions of the functional blocks shown in the block diagrams are mere examples, and thus a plurality of functional blocks may be implemented as a single functional block, or a single functional block may be divided into a plurality of functional blocks, or one or more functions may be moved to another functional block. Also, the functions of a plurality of functional blocks having similar functions may be processed by single hardware or software in a parallelized or time-divided manner.

(752) Also, the processing order of executing the steps shown in the flowcharts is a mere illustration for specifically describing the present disclosure, and thus may be an order other than the shown order. Also, one or more of the steps may be executed simultaneously (in parallel) with another step.

(753) A three-dimensional data encoding device, a three-dimensional data decoding device, and the

like according to one or more aspects have been described above based on the embodiments, but the present disclosure is not limited to these embodiments. The one or more aspects may thus include forms achieved by making various modifications to the above embodiments that can be conceived by those skilled in the art, as well forms achieved by combining structural components in different embodiments, without materially departing from the spirit of the present disclosure.

INDUSTRIAL APPLICABILITY

(754) The present disclosure is applicable to a three-dimensional data encoding device and a three-dimensional data decoding device.

Claims

1. A three-dimensional data encoding method, comprising: quantizing a first component using a first quantization parameter and quantizing a second component using a second quantization parameter, the first component and the second component configuring first attribute information among attribute information of each of three dimensional points; and generating a bitstream including the first component quantized, the second component quantized, the first quantization parameter, and a first difference between the first quantization parameter and the second quantization parameter, wherein in the quantizing using the first quantization parameter, the first component of at least one three-dimensional point included in one subspace of a current three-dimensional space is quantized further using a third quantization parameter, the one subspace being one of subspaces included in the current three-dimensional space, the current three-dimensional space including three-dimensional points to be processed, in the quantizing using the second quantization parameter, the second component of the at least one three-dimensional point is quantized further using a fourth quantization parameter, and in the generating, the bitstream generated further includes a second difference between the first quantization parameter and the third quantization parameter and a third difference between the second quantization parameter and the fourth quantization parameter.
2. The three-dimensional data encoding method according to claim 1, further comprising: quantizing a reflectance among the attribute information of each of the three-dimensional points, using a fifth quantization parameter, wherein in the generating, the bitstream generated further includes the reflectance quantized and the fifth quantization parameter.
3. The three-dimensional data encoding method according to claim 1, wherein in the generating, the bitstream generated further includes identification information indicating that the quantizing using the first quantization parameter was performed using the third quantization parameter and the quantizing using the second quantization parameter was performed using the fourth quantization parameter.
4. The three-dimensional data encoding method according to claim 1, further comprising: quantizing a third component using a sixth quantization parameter and quantizing a fourth component using a seventh quantization parameter, the third component and the fourth component configuring second attribute information among the attribute information of each of the three-dimensional points, wherein in the generating, the bitstream generated further includes the third component quantized, the fourth component quantized, the sixth quantization parameter, and a fourth difference between the sixth parameter and the seventh parameter.
5. A three-dimensional data decoding method, comprising: obtaining a quantized first component, a quantized second component, a first quantization parameter, and a first difference between the first quantization parameter and a second quantization parameter, by obtaining a bitstream; calculating a first component by inverse-quantizing the quantized first component using the first quantization parameter; and calculating a second component by inverse-quantizing the quantized second component using the second quantization parameter obtained from the first quantization parameter and the first difference, wherein the first component and the second component configure first

attribute information of three-dimensional points, wherein in the obtaining, a second difference between the first quantization parameter and a third quantization parameter and a third difference between the second quantization parameter and a fourth quantization parameter are further obtained by obtaining the bitstream, in the calculating of the first component, a first component of at least one three-dimensional point is calculated by inverse-quantizing the quantized first component using the first quantization parameter and the third quantization parameter obtained from the second difference, the at least one three-dimensional point being included in one subspace of a current three-dimensional space, the one subspace being one of subspaces included in the current three-dimensional space, the current three-dimensional space including three-dimensional points to be processed, the quantized first component being the luminance obtained by quantizing the first component of the at least one three-dimensional point using the first quantization parameter and the third quantization parameter, and in the calculating of the second component, a second component of the at least one three-dimensional point is calculated by inverse-quantizing the quantized second component using the second quantization parameter and the fourth quantization parameter obtained from the third difference, the quantized second component being the chrominance obtained by quantizing the second component of the at least one three-dimensional point using the second quantization parameter and the fourth quantization parameter.

6. The three-dimensional data decoding method according to claim 5, wherein in the obtaining, a quantized reflectance and a fifth quantization parameter are further obtained by obtaining the bitstream, the three-dimensional data decoding method further comprising: calculating a reflectance of the three-dimensional points by inverse-quantizing the quantized reflectance using the fifth quantization parameter.

7. The three-dimensional data decoding method according to claim 5, wherein in the obtaining, identification information indicating that quantizing was performed using the third quantization parameter and the fourth quantization parameter is further obtained by obtaining the bitstream, in the calculating of the first component, when the identification information indicates that quantizing was performed using the third quantization parameter and the fourth quantization parameter, the quantized first component is determined to be the luminance obtained by quantizing the first component of the at least one three-dimensional point, and in the calculating of the second component, when the identification information indicates that quantizing was performed using the third quantization parameter and the fourth quantization parameter, the quantized second component is determined to be the chrominance obtained by quantizing the second component of the at least one three-dimensional point.

8. The three-dimensional data decoding method according to claim 6, wherein in the obtaining, a quantized third component, a quantized fourth component, a sixth quantization parameter, and a fourth difference between the sixth quantization parameter and a seventh quantization parameter is further obtained by obtaining the bitstream, the three-dimensional data decoding method further comprising: calculating a third component by inverse-quantizing the quantized third component using the sixth quantization parameter; and calculating the fourth component by inverse-quantizing the quantized fourth component using the seventh quantization parameter obtained from the sixth quantization parameter and the fourth difference, wherein the third component and the fourth component configure second attribute information of the three-dimensional points.

9. A three-dimensional data encoding device, comprising: a processor; and a memory, wherein using the memory, the processor: quantizes a first component using a first quantization parameter and quantizes a second component using a second quantization parameter, the first component and the second component configuring first attribute information among attribute information of each of the three dimensional points; and generates a bitstream including the first component quantized, the second component quantized, the first quantization parameter, and a first difference between the first quantization parameter and the second quantization parameter, wherein in the quantizing using the first quantization parameter, the first component of at least one three-dimensional point

included in one subspace of a current three-dimensional space is quantized further using a third quantization parameter, the one subspace being one of subspaces included in the current three-dimensional space, the current three-dimensional space including three-dimensional points to be processed, in the quantizing using the second quantization parameter, the second component of the at least one three-dimensional point is quantized further using a fourth quantization parameter, and in the generating, the bitstream generated further includes a second difference between the first quantization parameter and the third quantization parameter and a third difference between the second quantization parameter and the fourth quantization parameter.

10. A three-dimensional data decoding device, comprising: a processor; and a memory, wherein using the memory, the processor: obtains a quantized first component, a quantized second component, a first quantization parameter, and a first difference between the first quantization parameter and a second quantization parameter, by obtaining a bitstream; calculates a first component by inverse-quantizing the quantized first component using the first quantization parameter; and calculates second component by inverse-quantizing the quantized second component using the second quantization parameter obtained from the first quantization parameter and the first difference, wherein the first component and the second component configure first attribute information of three-dimensional points, wherein in the obtaining, a second difference between the first quantization parameter and a third quantization parameter and a third difference between the second quantization parameter and a fourth quantization parameter are further obtained by obtaining the bitstream, in the calculating of the first component, a first component of at least one three-dimensional point is calculated by inverse-quantizing the quantized first component using the first quantization parameter and the third quantization parameter obtained from the second difference, the at least one three-dimensional point being included in one subspace of a current three-dimensional space, the one subspace being one of subspaces included in the current three-dimensional space, the current three-dimensional space including three-dimensional points to be processed, the quantized first component being the luminance obtained by quantizing the first component of the at least one three-dimensional point using the first quantization parameter and the third quantization parameter, and in the calculating of the second component, a second component of the at least one three-dimensional point is calculated by inverse-quantizing the quantized second component using the second quantization parameter and the fourth quantization parameter obtained from the third difference, the quantized second component being the chrominance obtained by quantizing the second component of the at least one three-dimensional point using the second quantization parameter and the fourth quantization parameter.
